



(AMARAVATI, AMRITAPURI, BANGALORE, COIMBATORE,
CHENNAI)

B.Tech in Artificial Intelligence and Data Science

Syllabus, 2023

PROGRAMME SPECIFIC OUTCOMES

- PSO1** Develop AI systems using a variety of algorithms and techniques, such as machine learning, natural language processing, computer vision, and robotics, to solve real-world problems and demonstrate proficiency in coding and implementing AI algorithms.
- PSO2** Analyze the performance of AI models and systems using appropriate evaluation metrics and techniques, and apply optimization methods to improve the efficiency and effectiveness of AI algorithms, while considering factors such as accuracy, speed, scalability, and resource utilization.
- PSO3** Develop ethical and responsible AI solutions by considering the ethical implications of AI technologies, including issues related to bias, fairness, transparency, privacy, and security, and propose strategies to mitigate potential ethical concerns in the development and deployment of AI systems.

SEMESTER 1

23xxxxxx	Mathematics for Intelligent Systems 1	L-T-P-C: 3- 0- 2- 4
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Course Objectives

- To introduce students to the fundamental concepts and techniques of linear algebra, ordinary differential equations, probability theory, complex numbers, and quantum computing that are necessary for further study in science and related fields.
- To enable students to apply the concepts they learn in practical situations by using analytical and numerical methods to model real-world problems.
- To expose students to the wide range of applications of linear algebra, ordinary differential equations, probability theory, complex numbers, and quantum computing within the scientific field and to inspire them to pursue further study or research in these areas.
- To introduce students to the fundamental concepts of quantum computing
- To develop students' ability to communicate mathematical concepts and solutions clearly and effectively.

Course Outcomes

After completing this course, students will be able to

CO1	Apply the fundamental concepts of linear algebra and calculus to solve canonical problems analytically and computationally
CO2	Model and simulate simple physical systems using ordinary differential equations
CO3	Apply the concept of probability and random variables to solve elementary problems
CO4	Explain the basic concepts of quantum computing and differentiate it from conventional computing

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	3	2	3	-	-	-	3	2	3	3	3	3	3
CO2	3	3	3	2	3	2		-	3	2	3	3	3	3	3
CO3	3	3	3	2	3	-	2	-	3	2	3	3	3	2	-
CO4	3	-	-	-	-	-	-	-	3	2	3	3	2	-	-

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Unit 1

Basics of Linear Algebra - Linear Dependence and independence of vectors - Gaussian Elimination - Rank of set of vectors forming a matrix - Vector space and Basis set for a Vector space – Vector spaces associated with a matrix, Dot product and Orthogonality -CR decomposition - Rotation matrices - Eigenvalues and Eigenvectors and its interpretation-Introduction to SVD-Computational experiments using Matlab/Excel/Simulink. How matrices arises in Science and Engineering: Linear system of equations, Rotation, Scaling and Translation in Computer Graphics, Neural Net computation, Markov chain associated with brand switching in Marketing Science etc.

Unit 2

Ordinary Linear differential equations, formulation - concept of slope, velocity and acceleration - analytical and numerical solutions- Impulse Response computations- converting higher order into first order equations - examples of ODE modelling in falling objects, satellite and planetary motion, Electrical and mechanical systems- Introduction to solving simple differential equations with Simulink- Introduction to one variable optimization - Taylor series- Computational experiments using Matlab /Excel/Simulink.

Unit 3

Introduction to random variables (continuous and discrete), mean, standard deviation, variance, sum of independent random variable, convolution, sum of convolution integral, probability distributions.

Unit 4

Introduction to quantum computing, Quantum Computing Roadmap, Quantum Mission in India, A Brief Introduction to Applications of Quantum computers, Quantum Computing Basics, Bracket Notation, Inner product, outer product, concept of state.

Text Books / References

Gilbert Strang, Introduction to Linear Algebra, Fifth Edition, Wellesley-Cambridge Press, 2016.

Gilbert Strang, Linear Algebra and Learning from Data, Wellesley, Cambridge press, 2019.

William Flannery, Mathematical Modelling and Computational Calculus, Vol-1, Berkeley Science Books, 2013.

Stephen Boyd and Lieven Vandenberghe, Introduction to Applied Linear Algebra – Vectors, Matrices, and Least Squares, 2018.

Bernhardt, Chris. Quantum computing for everyone. Mit Press, 2019. (From pages 1 to 36).

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 3)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Computational Mechanics 1 (Introduction to Dynamics/statics)	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- This course aims to introduce students to the fundamental concepts of computational mechanics, with a focus on developing computational models for mechanical systems using numerical methods.
- This course aims to provide students with a thorough understanding of kinematics, statics, and kinetics and their application to mechanical systems.
- This course aims to equip students with the skills and knowledge necessary to analyze the behavior of mechanical systems using computational mechanics tools and techniques.

Course Outcomes

After completing this course, students will be able to

CO1	Apply numerical methods to develop computational models for mechanical systems and analyze their behavior
CO2	Derive constitutive relations for mechanical systems in motion or at rest, including particles and rigid bodies, and use these equations to solve real world problems.
CO3	Evaluate the results of computational simulations and use this information to make informed decisions about mechanical systems design and optimization
CO4	Use software tools for computational mechanics, including code for solving equations of motion and simulating mechanical systems

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	1	3	-	-	1	1	2	-	2	-	-	-
CO2	3	3	1	1	3	-	-	1	2	3	-	2	-	-	-
CO3	3	3	3	2	3	-	-	1	3	3	-	2	2	1	1
CO4	3	3	3	2	3	-	-	1	3	3	-	2	2	2	2

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Unit 1 Kinematics and Statics

Position, velocity, and acceleration of particles, Newton's laws of motion, Work and energy, Rigid body kinematics, Translations and Rotations, Alternate representations of Rigid body Rotation - Rotation matrices, Euler angles, Axis-angle representations, Quaternions. Introduction to statics and equilibrium, Free body diagrams, Equilibrium of particles and rigid bodies, Computational aspects of solving kinematics and statics problems of real world systems.

Unit 2 Introduction to Kinetics

Cross product of two vectors, Inertial and Non-Inertial frame of reference, Linear momentum, Center of mass, Coriolis, Inertial and Centripetal forces, Acceleration in polar coordinates, Angular velocity, Angular momentum and Torque on particles, Computational aspects of solving kinetics problems of particles.

Unit 3 Kinetics of Rigid Bodies

Two particle system angular momentum, Inertia matrix, Moment and product of inertia, Principal axes theorem, Principal axes as eigenvector of Inertia matrix, Parallel axes theorem, Computational aspects of solving kinetics problems of particles, Introduction to Euler-Lagrange and Newton-Euler equations for solving rigid body dynamics. Euler-Lagrange equation derivation using one dimensional point mass example, Application of Euler-Lagrange equation for solving dynamics of simple mechanical systems.

Text Books / References

"Introduction to Computational Mechanics" by B. S. Choo and S. H. Han - 2005, 1st edition

"Engineering Mechanics: Dynamics" by J.L. Meriam and L.G. Kraige - 2016, 8th edition

"Engineering Mechanics: Statics" by J.L. Meriam and L.G. Kraige - 2016, 8th edition

"Vector Mechanics for Engineers: Statics and Dynamics" by Ferdinand P. Beer and E. Russell Johnston Jr. - 2015, 11th edition

"Mechanics of Materials" by James M. Gere and Barry J. Goodno - 2018, 9th edition

"Introduction to Classical Mechanics: With Problems and Solutions" by David Morin - 2008, 1st edition

Engineering Mechanics: Statics and Dynamics" by Irving H. Shames –2002, 4th edition.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Computational Thinking	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- Provide an insight on the importance of computational thinking
- Help to develop skills to solve problems using spreadsheet and MATLAB
- Provide logical thinking capabilities to solve problems .

Course Outcomes

After completing this course, students will be able to

CO1	Develop critical thinking
CO2	Apply logical thinking to solve problems
CO3	Develop skills to use spreadsheet for problem solving
CO4	Develop skills to use MATLAB for problem solving

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	2	3	-	2	2	-	-	-	-	-	-	-	1	2	1
CO2	2	3	-	2	2	-	-	-	-	-	-	-	1	2	1
CO3	2	3	-	2	2	-	-	-	-	-	-	-	1	2	1
CO4	2	3	-	2	2	-	-	-	-	-	-	-	1	2	1

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Unit 1

Computational Thinking, critical thinking, data representation, abstraction, decomposition- breaking problems into parts, basic data types, pseudocode, algorithms-methods to solve the problems, brute-force or exhaustive search problems, divide and conquer problems

Unit 2

Computational Thinking using spreadsheets, basic operations, cell references – relative and absolute, lookup operations, implement fractals – newton, Sierpinski triangle, L-system fractals, solve calculus based problems using spreadsheet, using spreadsheet for solving probability related problems

Unit 3

Computational thinking using matlab, basic operations, plotting of vectors, array and matrix operations, implement fractals – newton, Sierpinski triangle, L-system fractals, solve calculus based problems using matlab, using matlab for solving probability related problems

Text Books / References

Ferragina P, Luccio F. *Computational Thinking: First Algorithms, Then Code*. Springer; 2018

Beecher K. *Computational Thinking: A beginner's guide to Problem-solving and Programming*. BCS Learning & Development Limited; 2017.

Irfan Turk, *Matlab programming*, 2018

Noreen Brown, *Beginning Excel 2019*, 2019 .

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Elements of Computing - 1	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- The course will introduce the principles of number system conversions, Boolean logic, logic gates, and Boolean algebra.
- The course will aid the students in the design and analysis of combinational and sequential logic circuits.
- The course will also equip students to build a general-purpose computing system using elementary NAND gates through a simulation software

Course Outcomes

After completing this course, students will be able to

CO1	Demonstrate proficiency in performing number system conversion, manipulating Boolean Algebra expressions and realization of basic gates .
CO2	Implement different combinational logic circuits.
CO3	Implement different sequential logic circuits
CO4	Build a general-purpose computer using elementary NAND gates through a simulation software

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	2	3	-	-	-	3	2	3	3	1	1	1
CO2	3	3	3	3	3	2	1	1	3	2	3	3	2	2	2
CO3	3	2	3	3	3	2	1	1	3	2	3	3	2	2	2
CO4	3	2	3	2	3	-	-	-	3	2	3	3	2	2	2

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Unit 1

Number System, Conversions, Signed and Unsigned Binary Number Representation, Boolean algebra and Karnaugh Maps, Logic gates, Realization of basic gates using universal gates, Boolean function synthesis, Introduction to Hardware simulator platform Nand2teris, Hardware description language, Implementation of basic gates and its multi-bit and multiway versions in Nand2teris software suite.

Unit 2

Combinational Logic, Half Adder, Full Adder, Multiplexer and demultiplexer, Multi-bit and Multiway versions, Realization of Boolean functions using combinational logic, Arithmetic logic unit (ALU)- specification, design, Sequential logic, Flip Flops, Registers, RAM, ROM.

Unit 3

Von-Neumann architecture, Program Counter, Central Processing unit, Data Memory, Hack machine language specifications/ instructions for CPU design, Hack CPU Design, CPU Control logic, building a Hack Computer.

Text Books / References

Noam Nisan and Shimon Schocken, "Elements of Computing Systems", MIT Press, 2012.

M. Morris Mano, "Digital Design", 5th Edition, Pearson Education (Singapore) Pvt. Ltd., New Delhi, 2014.

John. M Yarbrough, "Digital Logic Applications and Design", Thomson Learning, 2006.

Anil K. Maini, "Digital Electronics", Wiley, 2014.

Thomas L. Floyd, "Digital Fundamentals", 10th Edition, Pearson Education Inc, 2011.

Donald D. Givone, "Digital Principles and Design", TMH, 2003.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Introduction to Electrical Engineering	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- To develop the understanding of both AC and DC circuits.
- To acquire the knowledge of circuit analysis techniques for solving both AC and DC circuits
- To learn state space modelling and solution of DC circuits.
- To establish connections between the concepts of electrical engineering, mathematics, and computational solution methods.

Course Outcomes

After completing this course, students will be able to

CO1	Apply and analyse circuit laws and the analysis techniques for solving electric circuits
CO2	Comprehend the transient behaviour of DC circuits and generate solutions computationally
CO3	Use the concepts of Thevenin and Norton equivalent networks to simplify complex circuits
CO4	Computationally solve the state space model of electric circuits

CO-PO Mapping

PO/	PO	PO	PO	PO	PO	PO	PO	PO	PO	PO	PO	PO	PSO	PSO	PSO
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PSO	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO															
CO1	3	3	2	2	3	1	-	-	3	2	3	3	2	1	1
CO2	3	3	3	3	3	2	-	-	3	2	3	3	2	3	2
CO3	3	2	3	3	3	2	2	-	3	2	3	3	2	2	3
CO4	3	3	3	2	3	2	2	-	3	2	3	3	3	3	2

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Unit 1 DC Circuit

EMF, Charge, Voltage, Current - Linear circuit elements – Energy and power - Ohms law – Kirchhoff's voltage and current law – Series parallel combination of R, L, C components – Voltage divider and current divider rules – Super position theorem – Nodal and Mesh Analysis – Step response of RL and RC Circuits (Transient behaviour) – Equivalent network: Thevenin and Norton.

Unit 2 AC Circuit

Impedance - Instantaneous, Average, Active, Reactive and Apparent Power – Power Factor – Phasors

Unit 3 Introduction to Control Systems

State Space Representation: State, State variable, and State Model – Canonical state space model for Series RLC Circuit – Solution using eigen values and eigen vectors.

Text Books / References

"Basic Electrical Engineering" by D. P. Kothari and J. Nagrath (4th edition), McGraw Hill, 2019.

"Electric Circuits" by James W. Nilsson and Susan Riedel (11th edition), Pearson, 2018.

"Modern Control Engineering" by Ogata (5th edition), Pearson, 2009.

"Linear Control Systems with MATLAB Applications" by B.S. Manke, Khanna, 12th edition, 2016 .

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Introduction to Biological Data	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- The course is aimed at educating students on fundamentals concept of biomolecules (sequence, structure, conformational state and functions) and the central dogma of biology.
- To teach them how to read /write molecular structure and sequence from specific file using scripting language (like awk, LINUX command vim) and introduced with LINUX system.
- It will explore different biological databases (Uniport, RSCB, GeneBank, PDBbind etc) and teach how to extract data from those databases.
- Ontology: Storing data in a structured manner through ontologies

- Teach fundamental concept to process the bio-signal and biomedical images.
- The fundamental concept of regression and classification will be introduced using example of biological data set.
- Introduced students with the biological visualization software (PyMol, VMD, ChemDraw, Arena3D etc.) and different webserver.

Course Outcomes

After completing this course, students will be able to

CO1	Develop ability to process (read, write, and analyse) biological data.
CO2	Analysis of different biological data
CO3	Design AI and ML research problem to address biological research problem
CO4	Develop knowledge to use different visualization tools and write script to handle software by command line

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	2	3	1	3	-	-	2	3	3	-	-	-
CO2	3	3	3	3	3	1	-	-	3	2	3	3	2	-	-
CO3	3	3	3	3	3	-	-	-	3	2	3	3	3	3	3
CO4	3	3	3	3	3	1	-	-	3	2	3	3	-	-	-

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Unit 1

Introduction to nucleic acid and protein sequence, structure, and function – introduction of drug molecules - Scripting language / Linux command to handle big biological data files (sequence and structure) – programming using python and R – Linux commands.

Unit 2

Explore different biological databases (RCSB, GenBank, DrugBank etc.) – introduction to protein family and ontology - protein functional database (Pfam, GO etc.) - extract data from database using scripting languages (awk, bash)

Unit 3

Biomolecular sequence descriptors – presents biomolecular structure using graph – quantify dynamics natures of biomolecules - scoring matrices to describe evolution relationship – fundamental of biomedical image analysis using python and MATLAB (CT scan, mammography, MRI etc.) – biomedical signal (ECG, EEG etc.) visualization and annotation using MATLAB -Virtualization software (PyMol, VMD etc) – TCL and Python programming to write code for PyMol and VMD.

Unit 4

Application of AI and machine learning to predict biological activity of biomolecules (regression) – AI-based biomolecular structure prediction models – Binary and multi classification problems related to biological data..

Text Books / References

“Introduction to Protein Structure” by Carl Ivar Branden, John Tooze

ECG Signal Processing, Classification and Interpretation: A Comprehensive Framework of Computational Intelligence" by Witold Pedrycz and Adam Gacek.

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	25
Quizzes (minimum 2)	Internal	20
Viva	Internal	10
Mid-Term Examination	Internal	15
Term Project/ End Semester Examination	External	30

CO1	Be introduced to the cultural ethos of Amrita Vishwa Vidyapeetham, and Amma's life and vision of holistic education
CO2	Understand the foundational concepts of Indian civilization like puruṣārtha-s, law of karma and varṇāśrama
CO3	Gain a positive appreciation of Indian culture, traditions, customs and practices
CO4	Imbibe spirit of living in harmony with nature, and principles and practices of Yoga
CO5	Get guidelines for healthy and happy living from the great spiritual masters

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															

CO1	-	-	-	-	-	3	2	3	-	-	-	2	-	-	-
CO2	-	-	-	-	-	3	1	3	-	-	-	2	-	-	-
CO3	-	-	-	-	-	3	1	3	-	-	-	2	-	-	-
CO4	-	-	-	-	-	3	3	3	-	-	-	2	-	-	-
CO5	-	-	-	-	-	3	1	3	-	-	-	2	-	-	-

Syllabus

Unit 1

Introduction to Indian culture; Understanding the cultural ethos of Amrita Vishwa Vidyapeetham; Amma's life and vision of holistic education.

Unit 2

Goals of Life – Purusharthas; Introduction to Varnasrama Dharma; Law of Karma; Practices for Happiness.

Unit 3

Symbols of Indian Culture; Festivals of India; Living in Harmony with Nature; Relevance of Epics in Modern Era; Lessons from Ramayana; Life and Work of Great Seers of India.

Text Books / References

Cultural Education Resource Material Semester-1

The Eternal Truth (A compilation of Amma's teachings on Indian Culture)

Eternal Values for a Changing Society. Swami Ranganathananda. BharatiyaVidyaBhavan.

Awaken Children (Dialogues with Mata Amritanandamayi) Volumes 1 to 9

My India, India Eternal. Swami Vivekananda. Ramakrishna Mission

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Periodical 1 (P1)	Internal	15
Periodical 2 (P2)	Internal	15
*Continuous Assessment (CA)	Internal	20
End Semester	External	50

*CA – Can be Quizzes, Assignment, Projects, and Reports.

22AVP103

Mastery Over Mind (MAOM)

L-T-P-C:2-0-2-3

Course Outcomes

After completing this course, students will be able to

CO1	To be able to describe what meditation is and to understand its health benefits
CO2	To understand the causes of stress and how meditation improves well-being.
CO3	To understand the science of meditation.
CO4	To learn and practice MA OM meditation in daily life.
CO5	To understand the application of meditation to improve communication and relationships.

CO6	To be able to understand the power of meditation in compassion-driven action.
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Unit 1: Describe Meditation and Understand its Benefits

A:Importance of meditation. How does meditation help to overcome obstacles in life (Pre-recorded video with Swami Shubhamritananda Puri)

Reading 1: Why Meditate? (Swami Shubamritananda ji)

MA OM Mastery Over Mind, School of Spiritual and Cultural Studies| Amritapuri Campus Course Outline, September 2022

Reading 2: 'Stillness of the Mind' Chapter 17 in Amritam Gamaya (2022). Mata Amritanandamayi Mission Trust.

Additional Reading: Abhyasa Yoga: The Yoga of Practice. (Br. Achyutamrita Chaitanya)

B:Understand how meditation works. Understand how meditation helps in improving physical and mental health. Understand how meditation helps in the development of personality (Pre-recorded video with Dr. Ram Manohar).

Unit 2: Causes of Stress and How Meditation Improves Well-being

A:Learn how to prepare for meditation. Understand the aids that can help in effectively practicing meditation. Understand the role of sleep, physical activity, and a balanced diet in supporting meditation. (Pre-recorded video with Dr. Ram Manohar)

Causes of Stress. The problem of not being relaxed. Effects of stress on health. How meditation helps to relieve stress. Basics of stress management at home and the workplace. (Pre-recorded video with Prof Udhaykumar)

Reading 1: Mayo Clinic Staff (2022, April 29). Meditation: A Simple, Fast Way to Reduce Stress. Mayo Clinic. <https://www.mayoclinic.org/tests-procedures/meditation/in-depth/meditation/art-20045858> (PDF provided)

Reading 2: 'Efficient Action.' Chapter 28 in Amritam Gamaya (2022). Mata Amritanandamayi Mission Trust.

Unit 3: The Science of Meditation

A:Learn how to prepare for meditation. Understand the aids that can help in effectively practicing meditation. Understand the role of sleep, physical activity, and a balanced diet in supporting meditation. (Pre-recorded video with Dr. Ram Manohar)

B:Causes of Stress. The problem of not being relaxed. Effects of stress on health. How meditation helps to relieve stress. Basics of stress management at home and the workplace. (Pre-recorded video with Prof Udhaykumar)

Reading 1: Mayo Clinic Staff (2022, April 29). Meditation: A Simple, Fast Way to Reduce Stress. Mayo Clinic. <https://www.mayoclinic.org/tests-procedures/meditation/in-depth/meditation/art-20045858> (PDF provided)

Reading 2: 'Efficient Action.' Chapter 28 in Amritam Gamaya (2022). Mata Amritanandamayi Mission Trust.

Unit 4: Practicing MA OM Meditation in Daily Life

Guided Meditation Sessions following scripts provided (Level One to Level Five)

Reading 1: MA OM and White Flower Meditation: A Brief Note (Swami Atmananda Puri)

Reading 2: 'Live in the Present Moment.' Chapter 71 in Amritam Gamaya (2022). Mata Amritanandamayi Mission Trust.

Unit 5: Improving Communication and Relationships

How meditation and mindfulness influence interpersonal communication. The role of meditation in improving relationship quality in the family, at the university and in the workplace. (Pre-recorded video with Dr Shobhana Madhavan)

Reading 1: Seppala E (2022, June 30th) 5 Unexpected Ways Meditation Improves

Relationships a Lot. Psychology Today.

<https://www.psychologytoday.com/intl/blog/feeling-it/202206/5-unexpected-ways-meditation-improves-relationships-lot>

Reading 2: 'Attitude.' Chapter 53 in Amritam Gamaya (2022). Mata Amritanandamayi Mission Trust.

Unit 6 Meditation and Compassion-driven Action

Understand how meditation can help to motivate compassion-driven action. (Pre-recorded video with Dr Shobhana Madhavan)

Reading 1: Schindler, S., & Friese, M. (2022). The relation of mindfulness and prosocial

behavior: What do we (not) know?. Current Opinion in Psychology, 44, 151-156.

Reading 2: 'Sympathy and Compassion.' Chapter 100 in Amritam Gamaya (2022). Mata Amritanandamayi Mission Trust.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Reflection	Internal	20
Group Activities	Internal	20
Class Participation	Internal	40
Written Examination	Internal	20

23xxxxxx	Technical Communication	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- To introduce the students to the fundamentals of mechanics of writing
- To facilitate them with the style of documentation and specific formal written communication
- To initiate in them the art of critical thinking and analysis
- To help them develop techniques of scanning for specific information, comprehension and organization of ideas
- To enhance their technical presentation skills.

Course Outcomes

After completing this course, students will be able to

CO1	To gain knowledge about the mechanics of writing and the elements of formal correspondence
CO2	To understand and summarise technical documents
CO3	To apply the basic elements of language in formal correspondence
CO4	To interpret and analyze information and to organize ideas in a logical and coherent manner
CO5	To compose project reports/ documents, revise them for language accuracy and make technical presentations

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO2	-	-	-	1	-	-	-	-	-	2	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	1	-	-	-	-	-	2	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	2	1	-	-	-	-	-

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Unit 1

Mechanics of Writing: Grammar rules -articles, tenses, auxiliary verbs (primary & modal) prepositions, subject-verb agreement, pronoun-antecedent agreement, discourse markers and sentence linkers

General Reading and Listening comprehension - rearrangement & organization of sentences

Unit 2

Different kinds of written documents: Definitions- descriptions- instructions-recommendations- user manuals - reports – proposals.

Formal Correspondence: Writing formal Letters Mechanics of Writing: impersonal passive & punctuation
Scientific Reading & Listening Comprehension

Unit 3

Technical paper writing: documentation style - document editing – proof reading - Organising and formatting
Mechanics of Writing: Modifiers, phrasal verbs, tone and style, graphical representation

Reading and listening comprehension of technical documents Mini Technical project (10 -12 pages)

Technical presentations

Text Books / References

Hirsh, Herbert. L “Essential Communication Strategies for Scientists, Engineers and Technology Professionals”. II Edition. New York: IEEE press, 2002

Anderson, Paul. V. “Technical Communication: A Reader-Centred Approach”. V Edition. Harcourt Brace College Publication, 2003

Strunk, William Jr. and White. EB. “The Elements of Style” New York. Alliyen& Bacon, 1999.

Riordan, G. Daniel and Pauley E. Steven. “Technical Report Writing Today” VIII Edition (Indian Adaptation). New Delhi: Biztantra, 2004.

Michael Swan. “ Practical English Usage”, Oxford University Press, 2000

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Periodical 1	Internal	10

Periodical 2	Internal	10
*Continuous Assessment (Theory) (CAT)	Internal	10
*Continuous Assessment (Lab) (CAL)	Internal	40
End Semester	External	30

*CA – Can be Quizzes, Assignment, Projects, and Reports.

SEMESTER 2

23xxxxxx	Mathematics for Intelligent Systems 2	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- To introduce students to the fundamental concepts of linear algebra, differential equations, optimization, and probabilistic modelling
- To enable students to apply the concepts they learn in practical situations by using analytical and numerical methods to model real-world problems.
- To expose students to the wide range of applications of linear algebra, ordinary differential equations, probability theory, and quantum computing within the scientific field and to inspire them to pursue further study or research in these areas.
- To equip students with advanced mathematical knowledge and problem-solving skills highly valued in various industries and research fields.

Course Outcomes

After completing this course, students will be able to

CO1	Implement matrix decomposition techniques to solve linear systems of equations.
CO2	Formulate optimization problems and solve them using gradient based and Newton's methods
CO3	Analyse data using fundamental techniques of probability.
CO4	Explain quantum entanglement, qubits and state vectors

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	3	2	3	-	-	-	3	2	3	3	3	3	3
CO2	3	3	3	2	3	2		-	3	2	3	3	3	3	3
CO3	3	3	3	2	3	-	2	-	3	2	3	3	3	2	-
CO4	3	-	-	-	-	-	-	-	3	2	3	3	2	-	-

Syllabus

Unit 1

Gaussian elimination – LU decomposition – Vector spaces associated with Matrices- Special orthogonal matrices - Fourier Series and Fourier Transform and its properties – Convolution - Projection matrix and

Regression - Convolution sum - Convolution Integral - Eigenvalues and Eigenvectors of Symmetric matrices - Eigenvalues and Eigen vectors of $A^T A$, $A A^T$ - Relationship between vector spaces associated with A , $A^T A$, $A A^T$ - Singular Value Decomposition – Concept of Pseudoinverse- Computational experiments using MATLAB/Excel/Simulink

Unit 2

Taylor series expansion of multivariate functions-conditions for maxima, minima and saddle points-Concept of gradient and Hessian matrices- Impulse Response computations- converting higher order into first order equations – concept of e^{At} - Multivariate regression and regularized regression -Theory of convex and non-convex optimization-Newton method for unconstrained optimization- Signal processing with regularized regression- Computational experiments using MATLAB/Excel/Simulink

Unit 3

Random variables and distributions - Expectation, Variance, Moments, Cumulants- Moment generating functions - Sampling from univariate distribution- various methods - Bayes theorem, Concept of Jacobian, and its use in finding pdf of functions of Random variables (RVs), Box-muller formula for sampling normal distribution - Concept of correlation and Covariance of two linearly related RVs.

Unit 4

Introduction to quantum computing–Introduction to spin – state vectors – Qubits – Entanglement. Measurement in Quantum Mechanics.

Textbooks / References

Gilbert Strang, Linear Algebra and Learning from Data, Wellesley, Cambridge press, 2019.

William Flannery, Mathematical Modelling and Computational Calculus, Vol-1, Berkeley Science Books, 2013.

Stephen Boyd and Lieven Vandenberghe, Introduction to Applied Linear Algebra – Vectors, Matrices, and Least Squares, 2018.

Douglas C. Montgomery and George C. Runger, Applied Statistics and Probability for Engineers, (2005) John Wiley and Sons Inc

Bernhardt, Chris. Quantum computing for everyone. Mit Press, 2019. (From pages 37 to 70).

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Computational Mechanics II	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- This course introduces students to the concepts and techniques of computational applied Mechanics, providing them with a solid foundation in analyzing and solving mechanical engineering problems using computational methods.
- This course equips students with the necessary skills to apply the Finite Element Method (FEM) and Computational Fluid Mechanics (CFD) for the analysis of common robotic applications.
- This course familiarizes students with the basic robotic control and its implementation for simple robotic systems.

Course Outcomes

After completing this course, students will be able to

CO1	Apply the Finite Element Method (FEM) to solve linear and nonlinear equations, demonstrating proficiency in utilizing FEM for computational analysis of mechanical engineering problems.
CO2	Analyze stress and strain analysis results, utilizing computational techniques to evaluate the structural behavior of beams and frames.
CO3	Perform computational fluid dynamics (CFD) simulations to analyze fluid flow over simple 3-dimensional objects. Implement reactive control systems for simple robotic systems.
CO4	Implement reactive control systems for simple robotic systems.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	3	2	3	-	-	-	3	3	-	2	-	-	-
CO2	3	3	3	3	3	-	-	-	3	3	-	2	1	-	-
CO3	3	3	3	2	3	-	-	-	3	3	1	2	2	1	-
CO4	3	3	3	2	3	-	-	-	3	3	-	2	2	2	1

Syllabus

Unit 1 Solid Mechanics

Stress and strain analysis, Stiffness matrices and element equations, Introduction to Finite Element Method (FEM), Discretization techniques: Meshing and Grid Generation, Solving linear and nonlinear equations using FEM, Finite element analysis software overview and usage, Modeling and analysis of beams and frames.

Unit 2 Fluid Mechanics

Introduction to fluids, Conservation of mass, Conservation of momentum, Navier-Stokes equations, Introduction to Computational fluid dynamics (CFD) simulations, Flow in channels & Flow over simple 3-dimensional objects simulations.

Unit 3 Introduction to Robotics Control

Chronology of Robotic control systems, Control objectives for robotic control, Model free or Reactive control – Proportional, Proportional Derivative, Proportional Integral, Proportional Integral Derivative controls. Computational aspects of reactive control for simple robotic systems.

Text Books / References

- “A First Course in the Finite Element Method” by Daryl L. Logan – 2016, CL Engineering; 6th edition.
- “MATLAB Guide to Finite Elements: An Interactive Approach” by Peter I. Kattan - 2007, Springer; 2nd edition.
- “Fluid Mechanics” by Frank M. White & Henry Xue – 2022, McGraw Hill.
- “Computational Fluid Dynamics: A Practical Approach” by Jiyuan Tu, Guan-Heng Yeoh & Chaoqun Liu – 2018, Butterworth-Heinemann; 3rd edition.

- Robot Modeling and Control by Mark W. Spong, Seth Hutchinson, and M. Vidyasagar -2005, 1st edition.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx

Object Oriented Programming

L-T-P-C: 2- 0- 2- 3

Course Objectives

- To introduce Objective Oriented Programming concepts.
- To equip the students to solve engineering problems by applying Object Oriented Concepts.
- To introduce development of GUI based applications.

Course Outcomes

After completing this course, students will be able to

CO1	Represent the problems using objects and classes.
CO2	Implement object-oriented concepts using the Java language
CO3	Apply object-oriented concepts to design and visualize programs using UML.
CO4	Implement applications using object-oriented features.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	2	3	-	-	-	2	2	-	3	1	2	1
CO2	3	3	3	3	3	-	-	-	2	2	-	3	1	2	1
CO3	3	2	3	3	3	-	-	-	2	2	-	3	1	2	1
CO4	3	2	3	3	3	-	-	-	2	2	-	3	1	2	1

Syllabus

Unit 1

Introduction to Java Language and Runtime Environment, JVM, Bytecode, Basic program syntax, Datatypes, Variables, Operators, Control statements, Loops, Arrays, Functions.

Unit 2

Object-oriented concepts- Abstraction, Encapsulation, Inheritance and Polymorphism. Class and objects, Constructor functions, Class members and methods, Class Instance variables, Garbage collector, Method overloading.

Unit 3

Unit 4

Text Books / References

John R. Hubbard, Schaum's Outline of Programming with Java, McGraw-Hill Education, 2004

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

CO1	Choose an appropriate data structure as applied to a specified problem
CO2	Use various techniques for representation of the data in the real world
CO3	Develop applications using data structures.
CO4	Test the logical ability for solving problems

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
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CO															
CO1	3	3	2	2	3	-	-	-	3	2	3	3	2	1	-
CO2	3	3	3	3	3	-	-	-	3	2	3	3	3	2	-
CO3	3	2	3	3	3	-	-	-	3	2	3	3	2	2	-
CO4	3	3	3	2	3	-	-	-	3	2	3	3	2	3	2

Syllabus

Unit 1

Data Structure – primitive and non-primitive, Array data structure, properties and functions, single and multi-dimensional arrays, simple problems, Basics of Algorithm Analysis, big-Oh notation, notion of time and space complexity, dynamic arrays

Unit 2

Linked List - singly linked list, doubly linked list, circular linked list- properties and functions, implementations, Sorting algorithms – selection, bubble, insertion, quick sort, merge sort, comparison of sorting algorithms, implementation using arrays.

Unit 3

Stack data structure, properties and functions, recursion, expression evaluation, Queue data structure - circular queue, double ended queue, properties, and functions

Unit 4

Binary Tree– arrays and linked list representation, tree traversals-preorder, postorder, inorder, level order. Graphs- directed and undirected graphs, adjacency list and matrices, Incidence matrices, path, graph traversals – breadth-first and depth-first, Shortest path- Dijkstra's algorithm, Bellman-Ford algorithm, Floyd-Warshall algorithm -

Text Books / References

Alfred V Aho, John E Hopcroft, Jeffrey D Ullman. Data Structures & Algorithms, Pearson Publishers, 2002.

'Maria Rukadikar S. Data Structures & Algorithms, SPD Publishers, 2011.

Michael T. Goodrich & Roberto Tamassia, Data Structures and Algorithms in Java, Wiley India Edition, Third Edition

Narasimha Karumanchi, Data Structures and Algorithms Made Easy in Java, CarrerMonk, 2011

Y. Langsam, M. Augenstin and A. Tannenbaum, Data Structures using C and C++, Pearson Education, 2002.

Lipschutz Seymour, Data Structures with C (Schaum's Outline Series), McGraw Hill Education India, 2004

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Elements of Computing Systems II	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- This course aims to provide an integrative, project-oriented approach to build software layers of a general-purpose computer system.
- The course will take the students through a series of software-layer construction tasks.
- This course will demonstrate how theoretical and applied techniques taught in other computer science courses are used in practice.

Course Outcomes

After completing this course, students will be able to

CO1	Develop and execute programs in low-level languages such as Hack machine language and assembly language
CO2	Create virtual machine specification or VM code for high-level and assembly languages
CO3	Develop programs in object-based language 'Jack'
CO4	Execute experiments related to basic concepts and functions of operating systems/compilers

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	2	3	-	-	-	3	2	3	3	1	1	1
CO2	3	3	2	2	3	-	-	-	3	2	3	3	2	2	1
CO3	3	2	3	2	3	-	-	-	3	2	3	3	2	2	2
CO4	3	2	2	2	3	-	-	-	3	2	3	3	2	2	2

Syllabus

Unit 1

Basic Computer architecture, Basic Elements of Machine Language, Hack Machine language, different sets of instructions, Hack programming, Assembly language vs Machine language, Assembler- assembly process, Hack assembly language, assembly language to binary conversions.

Unit 2

Virtual Machine I: Stack Arithmetic Background, VM Specification Part-1, Implementation and Perspective, Virtual Machine II: Program Control Background, VM Specification Part-2, Implementation, Perspective, High-Level Language: Background, The Jack Language Specification, Writing Jack Applications.

Unit 3

Compiler I - Syntax Analysis: Background, Specification, Implementation, Perspective. Compiler II – Code Generation: Background, Specification, Implementation, Perspective. Operating System: Background, the Jack OS Specification, Implementation, Perspective.

Text Books / References

Nisan, Noam, and Shimon Schocken. *The elements of computing systems: building a modern computer from first principles*. MIT press, 2005

M. Morris Mano *Computer System Architecture*, Prentice Hall, Third Edition.

Hennessy, John L., and David A. Patterson. *Computer architecture: a quantitative approach*. Elsevier, 5th Edition, 2011.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Introduction to Electronics	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- The course will lay down the basic concepts and techniques of electronics needed for advanced topics in AI.
- It will explore the concepts through computational/hardware experiments alongside introducing the concepts/theory behind it.
- It will help the students to perceive the engineering problems using the fundamental concepts in electronics.

Course Outcomes

After completing this course, students will be able to

CO1	Explain the basic concepts of analogy and digital electronics
CO2	Design and implement various electronic circuits using diodes and transistors and demonstrate an understanding of their applications in electronic devices and systems.
CO3	Design and implement operational amplifier circuits for a range of applications and demonstrate an understanding of their operation and behaviour in electronic systems
CO4	Model and analyse engineering problems and demonstrate the ability to propose and implement electronic solutions to these problems.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	2	3	-	-	-	3	2	3	3	1	1	1
CO2	3	3	3	3	3	-	-	-	3	2	3	3	2	2	2
CO3	3	3	3	3	3	-	-	-	3	2	3	3	2	2	2
CO4	3	2	3	3	3	-	-	-	3	2	3	3	3	3	3

Syllabus

Unit 1 Semiconductor Physics

Conduction in semiconductors, Doping, PN Junction, Semiconductor diodes: PN diodes, Zener diode, Rectifiers and filters, Clipping and clamping circuits, Voltage regulators.

Unit 2 Transistors

Bipolar Junction Transistor (BJT), Field Effect Transistors (FET), MOSFET, BJT amplifiers and oscillators, Transistor as a switch - Amplifiers: common amplifier configurations, voltage gain and current gain, input and output impedance, small signal analysis, frequency response, power amplifiers.

Unit 3 Operational Amplifiers (Op-Amps)

Ideal characteristics, inverting and non-inverting amplifiers, summing and difference amplifiers, integrators and differentiators, comparators, oscillators, Schmitt trigger, Multivibrators - Feedback Amplifiers - Oscillators - DAC and ADC

Text Books / References

“Microelectronics” by Jacob Millman and A. Grabel, Tata McGraw-Hill Publishers, Second Edition, New Delhi, 1999.

“Op-amps and Linear Integrated circuits” by Ramakant Gayakwad, , Prentice Hall, New Delhi, 1988.

‘Operational Amplifiers and Linear Integrated Circuits’ by Robert F. Coughlin and Frederick F. Driscoll, Pearson, 6th edition, 2000.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Introduction to Material Informatics	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- Provide a fundamental understanding of the field of material science and informatics, material properties.
- Explore the cutting-edge of modern material informatics tools, including machine learning, data analysis and visualization, and molecular/multiscale modelling.
- Learn how to work with small, sparse, and low-quality dataset.
- Analysis material failure and sustainability
- Develop AI-based computational model to design new materials with specific properties.

Course Outcomes

After completing this course, students will be able to

CO1	Apply modern material informatics tools including machine learning, simulations. Modelling, visualizations to solve a specific challenge most significantly.
CO2	Analysis material failure and sustainability
CO3	Design new materials and solve inverse design problem using AI.
CO4	Developed efficient predictive model using small, spares and low-quality dataset.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	3	3	2	-	-	2	2	1	2	-	3	3
CO2	3	3	1	3	3	2	2	1	2	3	2	2	-	3	-
CO3	3	3	3	3	3	2	2	1	3	3	2	2	3	3	3
CO4	3	3	3	3	3	2	2	1	3	3	2	2	3	2	3

Syllabus

Unit 1

Introduction to material science – structure, properties, and process spaces - process-structure-property linkages – foundation of material informatics – introduction to molecular mechanism and force field – quantification of dynamics properties of polymers (monte Carlo simulation, molecular dynamics simulation, normal mode analysis) – electronics structure of atoms (Gaussian, Gauss view, density functional theory)

Unit 2

Quantification and screening of materials properties - property prediction and optimization using AI - materials design and discovery using AI – how to handle small, sparsed, and low-quality dataset using AI.

Unit 3

Materials failure and sustainability analysis – new material and inverse materials design concept – solve inverse design using AI – enhance speed, efficacy and cost-effectiveness of material using AI - basic concept of quantum computing in material informatics.

Unit 4

Case studies of materials informatics (use of AI) in different fields (e.g. energy, aerospace, biomedical, etc.) - ethical considerations and limitations of materials informatics - future directions and challenges in materials informatics.

Text Books / References

“Material Informatics: Methods, Tools and Applications” by Olexandr Isayev, Alexander Tropsha and Stefano.

“Informatics for Materials Science and Engineering” by Krishna Rajan

“Machine Learning in Materials Informatics: Methods and Applications by Yuling An

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
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Assignments (minimum 2)	Internal	25
Quizzes (minimum 2)	Internal	20
Viva	Internal	10
Mid-Term Examination	Internal	15
Term Project/ End Semester Examination	External	30

23xxxxxx	Glimpses of Glorious India	L-T-P-C: 2- 0- 0- 2
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Course Objectives

- To deepen students' understanding and further their knowledge about the different aspects of Indian culture and heritage.
- To in still into students a dynamic awareness and understanding of their country's achievements and civilizing influences in various fields and at various epochs.

Course Outcomes

After completing this course, students will be able to

CO1	Get an overview of Indian contribution to the world in the field of science and literature.
CO2	Understand the foundational concepts of ancient Indian education system.
CO3	Learn the important concepts of Vedas and Yogasutra-s and their relevance to daily life.
CO4	Familiarize themselves with the inspirational characters and anecdotes from the Mahābhārata and Bhagavad Gita and Indian history.
CO5	Gain an understanding of Amma's role in the empowerment of women

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	-	-	-	-	-	3	3	-	-	-	-	2	-	-	-
CO2	-	-	-	-	-	1	-	3	-	-	-	2	-	-	-
CO3	-	-	-	-	-	3	3	3	-	-	-	2	-	-	-
CO4	-	-	-	-	-	3	3	3	-	-	-	2	-	-	-
CO5	-	-	-	-	-	1	-	1	-	-	-	-	-	-	-

Syllabus

Unit 1

To the World from India; Education System in India; Insights from Mahabharata; Human Personality. India's Scientific System for Personality Refinement.

Unit 2

The Vedas: An Overview; One God, Many Forms; Bhagavad Gita – The Handbook for Human Life; Examples of Karma Yoga in Modern India.

Unit 3

Textbooks / References

India's Rebirth. Sri Aurobindo. Auroville Publications

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Periodical 1	Internal	15
Periodical 2	Internal	15
*Continuous Assessment (CA)	Internal	20
End Semester	External	50

*CA – Can be Quizzes, Assignment, Projects, and Reports.

SEMESTER 3

23xxxxxx	Mathematics for Intelligent Systems 3	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- To provide students with advanced knowledge and skills in optimization, PDEs, probability and statistics, and quantum computing.
- To develop students proficiency in solving real-world problems in various domains, including physics, engineering, and computer science using the concepts of optimization, PDEs, and probability.
- To apply the concepts and techniques learned in the course to solve complex problems and communicate their solutions effectively to both technical and non-technical audiences.
- To equip students with advanced mathematical knowledge and problem-solving skills highly valued in various industries and research fields.

Course Outcomes

After completing this course, students will be able to

CO1	Apply the fundamental techniques of optimization theory to solve data science problems.
CO2	Analyse and solve computationally, physical systems using the formalism of partial differential equations.
CO3	Apply Markovian concepts in stochastic sequential systems.
CO4	Explain Bells Inequality and Quantum gates.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
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CO															
CO1	3	3	3	2	3	-	-	-	3	2	-	3	3	-	3
CO2	3	3	3	2	3	-	-	-	3	2	-	3	3	-	3
CO3	3	3	3	2	3	-	-	-	3	2	-	3	3	-	-
CO4	3	-	-	-	-	-	-	-	3	2	-	3	2	-	3

Syllabus

Unit 1

Direct methods for convex functions - sparsity inducing penalty functions- Constrained Convex Optimization problems - Krylov subspace -Conjugate gradient method - formulating problems as LP and QP - Lagrangian multiplier method-KKT conditions - support vector machines- solving by packages (CVXOPT) - Introduction to RKS - Introduction to DMD-Tensor and HoSVD- Linear algebra for AI.

Unit 2

Introduction to PDEs - Formulation and numerical solution methods (Finite difference and Fourier) for PDEs in Physics and Engineering- Computational experiments using Matlab/Excel/Simulink.

Unit 3

Multivariate Gaussian and weighted least squares - Markov chains - Markov decision Process

Unit 4

Introduction to quantum computing-Bells inequality-Quantum gates

Text Books / References

Gilbert Strang, Linear Algebra and Learning from Data, Wellesley, Cambridge press, 2019.

Gilbert Strang, "Differential Equations and Linear Algebra Wellesley", Cambridge press, 2018.

Stephen Boyd and Lieven Vandenberghe, Introduction to Applied Linear Algebra – Vectors, Matrices, and Least Squares, 2018.

Bernhardt, Chris. Quantum computing for everyone. Mit Press, 2019. (From pages 71 to 140).

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Modelling, Simulation & Analysis	L-T-P-C: 2- 0- 2- 4
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Course Objectives

- This course intends to provide students with the ability to create and analyze mathematical models of physical systems using various techniques such as bond graph modeling and system transfer functions.

- This course aims to equip students with the skills to use simulation tools such as MATLAB to simulate and analyze the behavior of dynamic systems and validate and verify simulation models.
- The course aims to develop student's abilities to apply system analysis and optimization techniques such as block diagram algebra, signal flow graphs, state variable formulation, frequency response, and Bode plot, to engineering problems.

Course Outcomes

After completing this course, students will be able to

CO1	Apply various modeling techniques including physical, mathematical, and computer-based modeling for engineering applications.
CO2	Analyze different system models, including basic ones such as mechanical, electrical, hydraulic, pneumatic, and thermal systems, and advanced ones such as electro-mechanical, hydro-mechanical, and robotic systems.
CO3	Apply various methods for modeling and simulating the behavior dynamic systems, including bond graph modeling, simulation using MATLAB, and parameter estimation methods.
CO4	Design simulations and analysis for engineering problems using optimization, block diagram algebra, signal flow graphs, state variable formulation, frequency response, and Bode plot.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	1	1	1	-	-	1	2	2	-	1	-	-	-
CO2	3	3	1	1	1	-	1	1	2	2	1	1	-	1	1
CO3	3	3	3	3	3	1	1	1	2	2	2	1	2	2	1
CO4	3	3	2	2	2	1	1	1	2	1	2	1	2	2	1

Syllabus

Unit 1

Introduction to modelling - Examples of models, Modelling of dynamic systems, Introduction to simulation - Matlab as a simulation tool, Bond graph modelling - Bond graph model and causality, Generation of system equation, Methods of drawing bond graph models - Mechanical systems, Electrical systems. Basic system models – Mechanical systems, Electrical systems, Hydraulic systems, Pneumatic systems, and Thermal systems

Unit 2

System models – Linearity and nonlinearity in systems, Combined rotary and translator systems, Electro-Mechanical systems, Hydro-mechanical systems, Robotic systems, Dynamic response of 1st and 2nd order systems, Performance measures for 2nd order system, System transfer functions – 1st and 2nd order systems

Unit 3

Block diagram algebra, Signal flow graphs, State variable formulation, Frequency response, Bode plot, Simulation using Matlab, Simulation of - simple and compound pendulums, planar mechanisms, and wheeled mobile robots, Validation and verification of simulation models, Parameter estimation methods, Parameter estimation examples, System identification, Introduction to optimization, Optimization with modeling engineering problems.

Text Books / References

Borutzky, Wolfgang, "Bond Graph Modelling of Engineering Systems", Springer, 2011.

Esfandiari, Ramin S. and Bei Lu, "Modeling and Analysis of Dynamic Systems", CRC Press, 2010.

Gardner, John F., Bohdan T. Kulakowski, and J. Lowen Shearer. "Dynamic Modeling and Control of Engineering Systems." Publisher: Cambridge University Press; 3rd edition (2008).

Karnopp, Dean C., Donald L. Margolis, and Ronald C. Rosenberg. "System Dynamics: Modeling, Simulation, and Control of Mechatronic Systems." Publisher: Wiley; 5th edition (2012).

Lennart, L. and Torkel, G., "Modeling of Dynamic Systems", Prentice Hall, 1994.

Woods, Robert L. and Kent L. Lawrence. "Modeling and Simulation of Dynamic Systems." Publisher: Prentice Hall; 1st edition (1997).

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Introduction to Robotics	L-T-P-C: 2- 0- 2- 4
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Course Objectives

- This course aims to provide students with an overview of concepts to applications of robotics.
- This course intends to equip students with the ability to design and analyze simple and elementary robotic systems (Up to 4 DOF systems like SCARA) using mathematical and computational tools.
- This course aims to give students elementary hands-on experience in programming robotic systems using Robotic Toolbox in python/MATLAB.

Course Outcomes

After completing this course, students will be able to

CO1	Explain facts pertaining to robotics including history, sub-fields, and applications.
CO2	Explain the elementary concepts required for modelling robotic systems.
CO3	Develop mathematical and mechanistic models for simple robotic systems.
CO4	Use tools such as Robotics toolbox to program and visualize simple robotic systems

CO-PO Mapping

PO	P O 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PS O3
CO															
CO1	1	1	1	1	1	1	1	-	3	3	-	2	-	-	-
CO2	2	2	1	1	2	-	-	-	2	2	-	2	1	-	-
CO3	2	2	2	1	2	-	-	-	2	2	-	2	2	1	-
CO4	2	2	1	1	3	-	-	-	2	2	-	2	1	1	-

Syllabus

Unit 1 Overview of Robotics

Definition and History of Robots, Applications of robots, Current trend in robotics, Basic mathematics for robotics – Vectors, Matrices and Linear Algebra concepts, Rigid body transformations – Translation and Rotation, Homogeneous Transformation matrix.

Unit 2 Kinematics of Simple Robotic Systems

Forward Kinematics of simple industrial robotic systems, Inverse kinematics of simple industrial robotic systems, Differential Kinematics of simple industrial robotic systems, Kinematics of simple wheeled mobile robots.

Unit 3 Dynamics and Control of Simple Robotic Systems

Introduction to rigid body kinetics, Euler-Lagrange equation of simple robotic systems, Forward and Inverse dynamics of simple robotic systems, Velocity based control of simple robotic systems, Torque based control of robotic systems.

Textbooks

Corke P. Robotics, Vision and Control. Springer; 2017.

Craig J J. Introduction to Robotics: Mechanics and Control. Pearson Publishing, 2017.

Spong M.W., Hutchinson S. and Vidyasagar M, Robot Modeling and Control, Wiley, 2006

Saha S K, Introduction to Robotics, McGraw Hill publishing, 2014

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Software-Defined Communication Systems	L-T-P-C: 2- 0- 2- 4
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Course Objectives

- Understand the basic principles of communication systems, including signal analysis, system characteristics, and different types of modulation/demodulation techniques.
- Develop practical skills in using SDR platforms and tools such as MATLAB Simulink, GNU Radio Companion, RTL-SDR, and ADALM Pluto to implement analog and digital modulation/demodulation techniques and analyse signals/spectra.

Course Outcomes

After completing this course, students will be able to

CO1	Analyse different signal attributes related to communication system
CO2	Design and implement basic analog communication techniques using software defined radio platforms
CO3	Design and implement basic digital communication techniques using software defined radio platforms

CO4	Develop an appreciation of the role of AI in communication systems
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CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	2	3	3	-	-	3	3	2	2	2	1	1
CO2	3	3	3	2	3	3	-	-	3	3	3	2	2	2	1
CO3	3	3	3	2	3	3	-	-	3	3	3	2	2	2	1
CO4	3	3	3	3	3	3	-	2	3	3	3	3	1	2	3

Syllabus

Unit 1

Introduction to communication systems, introduction to signals, different types of signals and their characteristics, concept of system, linear time-invariant (LTI) system, sinusoids- concept of frequency, in-phase and quadrature component, bandwidth, pass band and stop band, Introduction to SDR platforms and devices- MATLAB Simulink and GNU radio Companion (GRC), RTL-SDR and Adalm Pluto. Signal analysis/ spectrum analysis and visualization using SDR tools.

Unit 2

Need for modulation, analog modulation schemes, amplitude modulation (AM) and its types - AM-DSB-SC, AM-DSB-TC, SSB. AM Demodulation schemes, angle modulation- frequency modulation (FM) -Narrowband and wideband, phase modulation, FM demodulation, implementation of analog modulation/demodulation schemes using SDR tools.

Unit 3

Quadrature amplitude modulation and demodulation, pulse analog modulation schemes, digital carrier modulation/demodulation Schemes- amplitude shift keying (ASK), frequency shift keying (FSK), phase shift keying (PSK), M-ary signalling, BPSK, QPSK, implementation of digital modulation/demodulation schemes using SDR tools. Multicarrier modulation- OFDM, MIMO, Prospects of AI in communication system- radio signal or modulation classification.

Text Books / References

Wyglinski, Alexander M., Robin Getz, Travis Collins, and Di Pu. Software-defined radio for engineers. Artech House, 2018.

QasimChaudhari, Wireless Communications from the Ground Up: An SDR Perspective, 2018

Andrew Barron, Software Defined Radio: for Amateur Radio Operators and Shortwave Listeners, 2019

C.R. Johnson and W.A. Sethares, Software Receiver Design: Build Your Own Digital Communication System in Five Easy Steps, Cambridge University Press, 2011

Proakis, John G., Masoud Salehi, and Gerhard Bauch. *Contemporary communication systems using MATLAB*. Cengage Learning, 2012.

Wygłinski, Alexander M., and Di Pu. *Digital communication systems engineering with software-defined radio*. Artech House, 2013.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (Minimum 3)	Internal	30
Quizzes (Minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	20

23xxxxxx	Advanced Data Structures & Algorithm Analysis	L-T-P-C: 2- 0- 2- 4
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Course Objectives

- To impart various design techniques for formulation of algorithm.
- To understand basic categories of algorithms.
- To comprehend basic complexity classes.
- To acquaint with will know tractable and intractable problems and map solutions to it.

Course Outcomes

After completing this course student will be able to,

CO1	Develop skills for analyzing algorithmic strategies
CO2	Analyse and apply appropriate algorithmic technique for a given problem
CO3	Implementing standard algorithms on arrays, strings, trees and graphs
CO4	Visualize multidimensional geometry of data structure and concurrency.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	3	3	3	3	1	-	3	3	2	3	3	1	1
CO2	3	3	3	2	3	2	-	-	3	3	2	3	3	2	2
CO3	3	3	3	3	2	1	-	-	3	3	3	3	3	3	3
CO4	3	3	3	3	2	1	-	-	3	3	3	3	2	3	3

Syllabus

Unit 1

Full binary tree, complete binary tree, perfect binary tree, balanced binary tree, binary search tree, properties and functions

Unit 2

Binary Heap Data Structure-Heap property, properties and functions, Heapsort, AVL Tree – balance factor, rotating the subtrees in an AVL tree –right rotation, left rotations, left-right and right-left rotate, operations on AVL trees- insertion and deletion

Unit 3

Trie data structure- basic operations, simple problems, Hashing and Hash Tables –hash functions, collision, collision avoidance methods, Merkel trees

Unit 4

Fundamentals of the Analysis of Algorithmic Efficiency –Asymptotic Notations and growth rate-Empirical Analysis - Divide and Conquer Methodology - Dynamic programming - Knapsack Problem and Memory functions. Greedy Technique - Iterative methods – Backtracking - Branch and Bound

Text Books / References

Mehlhorn, Kurt, Peter Sanders, and Peter Sanders. *Algorithms and data structures: The basic toolbox*. Vol. 55. Berlin: Springer, 2008.

Bhim P Upadhyaya, *Data Structures and Algorithms with Scala*. Springer International Publishing, 2019.

Aho, Alfred V. "Data Structures and Algorithms, Addison-Wesley." Reading, Mass. (1983).

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. 2009. *Introduction to Algorithms, Third Edition (3rd ed.)*. The MIT Press

Jeffrey McConnell, *Analysis of algorithms*. Jones & Bartlett Publishers, 2nd Revised edition, 2007

Anany Levitin, *Introduction to the Design and Analysis of Algorithms, Third Edition*, Pearson Education, 2012

Harsh Bhasin, *Algorithms Design and Analysis*, Oxford university press, 2016

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Introduction to AI and Machine Learning	L-T-P-C: 2- 0- 2- 4
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Course Objectives

- To introduce fundamentals of AI.
- To introduce fundamentals of Data Science.
- To introduce different tools and techniques used in AI and Data Science.

Course Outcomes

After completing this course, students will be able to

CO1	Analyse different elements of an AI system.
CO2	Analyse different types of data representation.
CO3	Apply concepts of AI and Data Science to solve canonical problems.
CO4	Implement basic computational tools pertinent to AI and Data Science to solve canonical problems.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	2	2	2	3	2	2	2	2	2	-	2	3	2	3
CO2	2	2	2	2	3	-	-	-	2	2	-	2	3	2	3
CO3	2	2	2	2	3	-	-	-	2	2	-	2	3	2	3
CO4	3	2	2	2	3	-	-	-	2	2	2	2	3	2	3

Syllabus

Unit 1

History and Foundations of AI and Data Science, Applications of AI and Data Science, Career paths pertinent to AI and Data Science

Unit 2

Rational Intelligent Agents, Agents and Environments, Nature of Environments, Structure of Agents. Introduction - Overview of Data Science – Introduction to Statistics: Sampling, Sample Means and Sample Sizes - Descriptive statistics: Central tendency, dispersion, variance, covariance, kurtosis, five-point summary

Unit 3

Basic tools for AI and Data Science, Introduction to Data Science process pipeline, Different representations of Data, Importance of pre-processing the data, Elementary Applications of AI and Data Science

Text Books / References

Russell, Stuart Jonathan, Norvig, Peter, Davis, Ernest. Artificial Intelligence: A Modern Approach. United Kingdom: Pearson, 2010.

Deepak Khemani. A First Course in Artificial Intelligence. McGraw Hill Education (India), 2013.

Denis Rothman. Artificial Intelligence by Example, Packt, 2018.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

Course Objective

- This course helps students to understand the fundamental networking concepts and standards.
- This course helps students to understand the function of TCP/IP layers and the protocols involved.
- This course helps students to understand the configuration of different networks and routing using simulator/emulator.
- This course helps students to understand the importance and application of artificial intelligence in computer networks
- This course gives an introduction to the concepts of software defined networks and its applications.

Course Outcomes

After completing this course, students will be able to

CO1	Analyze the requirements for a given organizational structure to select the most appropriate networking architecture and technologies.
CO2	Analyze the working of protocols in the internet protocol stack for network applications.
CO3	Configure a router using simulator/emulator.
CO4	Analyze the network data to detect potential security threats in a network

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	-	-	-	1	2	1	-	1	-	-	-	1	-	2	-
CO2	2	2	2	2	2	-	-	1	-	-	-	2	3	2	1
CO3	2	2	2	2	2	-	-	1	-	-	-	2	3	2	1
CO4	1	1	1	2	2	1	-	1	-	-	-	2	1	-	1

Syllabus

Unit 1

Basic concepts of computer networks, Internet- network edge, network core, delay, loss, and throughput in packet switched networks, network topology, types of networks, Internet standards and organization. OSI layer stack, protocols in the context of the Internet protocol stack. Introduction to AI powered networks that monitor the connected devices and their bandwidth requirements

Unit 2

Application Layer – Protocols in Web and Email applications, Peer-to-Peer Applications. Transport Layer – connection-oriented and connectionless service, protocols, and socket programming. Network Layer – Internet Protocol, Host Addressing for subnets, Routing and Forwarding principles. Data link and Physical layer concepts for wired and wireless network

Unit 3

Network security- analyze the network traffic. Introduction to Software Define Networks

Text Books / References

Kurose, James F. *Computer networking: A top-down approach featuring the internet*, 3/E. Pearson Education India, 2005.

Andrew, S. *"Tanenbaum-Computer Networks –Prentice Hall."* New Jersey (2003).

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Leadership from Ramayana	L-T-P-C: 2- 0- 2- 3
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Syllabus

Chapter 1 - Introduction to the Great Itihasa

Chapter 2 - Bala-Kāṇḍa: (Preparing for the renowned mission.)

And Ayodhya-Kāṇḍa: (Harbinger of an Entire Tradition of Nobleness.)

Chapter 3 - Araṇya-Kāṇḍa: (Tale of the forest life)

And Kishkindha-Kāṇḍa: (The Empire of Holy Monkeys.)

Chapter 4 - Sundara-Kāṇḍa: (Heart of the Ramayana)

And Yuddha-Kāṇḍa: (The most popular part of the Ramayana)

Chapter 5 - Ramayana and Modern-day learning

Chapter 6 - Ecological Awareness in the Ramayana

Chapter 7 - Different Ramayana: (Epic that connects the world)

Chapter 8 - Uttara-Kāṇḍa: (An attempt to explain the untold stories)

Text Books / References

1. *Valmiki Ramayana* (English translation published by Gita Press)
2. *Skanda Purana*
3. *Ramayana - C Rajagopalachari* .

23xxxxx	Soft Skills	L-T-P-C: 2-0-2-2
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Course Outcomes

After completing this course, students will be able to

CO1	Soft Skills:At the end of the course, the students would have developed self-confidence and positive attitude necessary to compete and challenge themselves. They would also be able to analyse and manage their emotions to face real life situations.
CO2	Soft Skills:At the end of the course, the students would hone their presentation skills by understanding the nuances of content creation, effective delivery, use of appropriate body language and the art of overcoming nervousness to create an impact in the minds of a target

	audience.
CO3	Aptitude: At the end of the course, the student will have acquired the ability to analyze, understand and classify questions under arithmetic, algebra and logical reasoning and solve them employing the most suitable methods. They will be able to analyze, compare and arrive at conclusions for data analysis questions.
CO4	Verbal: At the end of the course, the students will have the ability to dissect polysyllabic words, infer the meaning, inspect, classify, contextualise and use them effectively.
CO5	Verbal: At the end of the course, the students will have the ability to understand the nuances of English grammar and apply them effectively..
CO6	Verbal: At the end of the course, the students will have the ability to identify, analyse and interpret relationship between words and use the process of elimination to arrive at the answer. They will also have the ability to judge, evaluate, summarise, criticise, present and defend their perceptions convincingly.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	-	-	-	-	-	-	-	2	3	3	-	3	-	-	-
CO2	-	-	-	-	-	-	-	-	2	3	-	3	-	-	-
CO3	-	3	-	2	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	3	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	3	-	-	-
CO6	-	-	-	-	-	-	-	-	-	3	-	3	-	-	-

Syllabus

Soft skills and its importance: Pleasure and pains of transition from an academic environment to work - environment. Need for change. Fears, stress and competition in the professional world. Importance of positive attitude, Self-motivation and continuous knowledge upgradation.

Self-confidence: Characteristics of the person perceived, characteristics of the situation, characteristics of the perceiver. Attitude, values, motivation, emotion management, steps to like yourself, positive mental attitude, assertiveness.

Presentations: Preparations, outlining, hints for efficient practice, last minute tasks, means of effective presentation, language, gestures, posture, facial expressions, professional attire.

Vocabulary building: A brief introduction into the methods and practices of learning vocabulary. Learning how to face questions on antonyms, synonyms, spelling error, analogy, etc. Faulty comparison, wrong form of words and confused words like understanding the nuances of spelling changes and wrong use of words. Listening

skills: The importance of listening in communication and how to listen actively.

Prepositions, articles and punctuation: A experiential method of learning the uses of articles and prepositions in sentences is provided.

Problem solving level I: Number system; LCM &HCF; Divisibility test; Surds and indices; Logarithms; Ratio,proportions and variations; Partnership;

Problem solving level II: Time speed and distance; work time problems;

Data interpretation: Numerical data tables; Line graphs; Bar charts and Pie charts; Caselet forms; Mix diagrams; Geometrical diagrams and other forms of data representation.

Logical reasoning: Family tree; Deductions; Logical connectives; Binary logic; Linear arrangements; Circular and complex arrangement; Conditionalities and grouping; Sequencing and scheduling; Selections; Networks; Codes; Cubes; Venn diagram in logical reasoning; Quant based reasoning; Flaw detection; Puzzles; Cryptogarithms.

TEXTBOOKS /REFERENCE

A Communicative Grammar of English: Geoffrey Leech and Jan Svartvik. Longman, London.

Adair. J., (1986), "Effective Team Building: How to make a winning team", London, U.K: Pan Books.

Gulati. S., (2006) "Corporate Soft Skills", New Delhi, India: Rupa& Co.

The Hard Truth about Soft Skills, by Amazone Publication.

Quantitative Aptitude by R. S. Aggarwal, S. Chand

Quantitative Aptitude – Abijith Guha, TMH.

Quantitative Aptitude for Cat - Arun Sharma. TMH.

Books on GRE by publishers like R. S. Aggarwal, Barrons, Kaplan, The Big Book, and Nova.

More Games Teams Play, by Leslie Bendaly, McGraw Hill Ryerson.

The BBC and British Council online resources

Owl Purdue University online teaching resources

www.the grammarbook.com - online teaching resources www.englishpage.com- online teaching resources

SEMESTER 4

23xxxxxx	Mathematics for Intelligent Systems 4	L-T-P-C: 2- 0- 2- 4
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Course Objectives

- To provide students with advanced knowledge and skills in optimization, statistical estimation theory, and quantum computing.
- To understand and analyze special matrices used in various areas of signal processing and data analysis.
- To learn optimization techniques for convex and non-convex problems, and their application to machine learning problems.
- To introduce statistical estimation theory and hypothesis testing, and their relevance to data analysis.
- To provide an overview of quantum computing and its potential applications in various field

Course Outcomes

After completing this course, students will be able to

CO1	Apply proximal algorithms, augmented Lagrangian, and ADMM to solve convex and non-convex optimization problems.
CO2	Develop optimization algorithms used in neural networks.
CO3	Apply statistical estimation theory and hypothesis testing to data analysis applications.
CO4	Apply quantum computing concepts to solve problems in various fields including cryptography and optimization.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	3	2	3	-	-	-	3	2	2	3	3	-	3
CO2	3	3	3	2	3	-	-	-	3	2	2	3	3	-	3
CO3	3	3	3	2	3	-	-	-	3	2	2	3	3	-	3
CO4	3	3	3	1	3	-	-	2	3	2	2	3	2	-	3

Syllabus

Unit 1

Special Matrices: Fourier Transform, discrete and Continuous, Shift matrices and Circulant matrices, The Kronecker product, Toeplitz matrices and shift invariant filters, Hankel matrices, DMD and need of Hankelization - Importance of Hankelization – DMD and its variants - Linear algebra for AI

Unit 2

Matrix splitting and Proximal algorithms - Augmented Lagrangian- Introduction to ADMM, ADMM for LP and QP - Optimization methods for Neural Networks: Gradient Descent, Stochastic gradient descent- loss functions and learning functions

Unit 3

Basics of statistical estimation theory and testing of hypothesis.

Unit 4

Introduction to quantum computing- Bells's circuit, Superdense coding, Quantum teleportation. Programming using Qiskit, MATLAB.

Text Books / References

Gilbert Strang, Linear Algebra and Learning from Data, Wellesley, Cambridge press, 2019.

Gilbert Strang, "Differential Equations and Linear Algebra Wellesley", Cambridge press, 2018.

Stephen Boyd and, Lieven Vandenberghe, "Introduction to Applied Linear Algebra – Vectors, Matrices, and Least Squares", Cambridge University Press, 2018

Bernhardt, Chris. Quantum computing for everyone. Mit Press, 2019. (From pages 71 to 140).

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

Course Objectives

- This course provides the basic concepts of deep learning and implementation using Matlab/Python.
- This course provides the application of deep learning algorithms in signal and image data analysis.
- This course covers the concept of deep learning algorithms such as transfer learning and attention models for signal and image analysis.

Course Outcomes

After completing this course, students will be able to

CO1	Apply the fundamentals of deep learning.
CO2	Apply deep learning algorithms using MATLAB/Python.
CO3	Apply deep learning models for signal analysis
CO4	Implement deep learning models for image analysis.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO3
CO															
CO1	3	2	2	-	3	2	-	-	3	3	-	3	2	3	2
CO2	3	2	2	2	3	3	-	-	3	3	2	3	3	3	2
CO3	3	2	2	2	3	3	-	-	3	3	2	3	3	3	2
CO4	3	3	2	2	3	3	-	-	3	3	2	3	3	3	3

Syllabus

Unit 1

Introduction to neural networks – Gradient Descent Algorithm - Deep Neural Networks (DNN) –Convolutional Neural Network (CNN) – Recurrent Neural Network (RNN): Long-Short- Term-Memory (LSTM).

Unit 2

Pre-processing: Noise Removal using deep learning algorithms - Feature Extraction - Signal Analysis: Time Series Analysis, CNNs, Auto encoders.

Unit 3

Image Analysis: Transfer Learning, Attention models- Ensemble Methods for Signal and Image Analysis.

Textbooks & References:

Bishop C.M, "Pattern Recognition and Machine Learning", Springer, 1st Edition, 2006.

Goodfellow I, Bengio Y, Courville A, & Bengio Y, "Deep learning", Cambridge: MIT Press, 1st Edition, 2016.

Soman K.P, Ramanathan. R, "Digital Signal and Image Processing – The Sparse Way", Elsevier, 1st Edition, 2012.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (Minimum 2)	Internal	30
Quizzes (Minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Introduction to IoT	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- To provide hands-on experience in IoT concepts such as sensing, actuation, and communication.
- To develop program skills in Arduino and Raspberry-PI programming for IoT applications.
- To introduce the process of interfacing actuators and sensing devices to Arduino and Raspberry PI.
- To impart the knowledge of networking concepts that enable wired and wireless communication among devices for IoT applications.
- To introduce cloud platforms for storing and implementing IoT applications.

Course outcomes

Upon completion of the course, students will be able to,

CO1	Familiarize with the fundamental concepts of Internet of Things. .
CO2	Develop skills in programming and hardware platform like Arduino and Raspberry-PI for IOT applications.
CO3	Familiarize with the design and implementation of IOT protocols and connecting devices for IOT application.
CO4	Analyse and integrate the IOT applications to cloud service.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	2	2		1	1	1	1	-	-			1	1	1	1
CO2	2	3	2	2	2	1	2	-	-	-	1	2	2	3	1
CO3	2	3	1	3	2	1	1	-	-	-	1	2	1	2	1
CO4	2	3	2	2	2	1	2	-	-	-	1	2	1	2	2

Syllabus

Unit 1

Introduction to IOT Overview of machine-to-machine communication - Sensing – Actuators-Basics of Networking- Introduction to Micro-Controllers-Introduction to Embedded systems.

Unit 2

Basics of networking for device-to-device communication ,Communication Protocols – wired and wireless communication – Network Topology-Sensor Networks-Introduction to Arduino and Raspberry-PI-Introduction

to IOT protocols-MQTT-COAP-Wi-Fi and Bluetooth connections in Arduino-Raspberry-PI Ethernet and Wi-Fi connectivity

Unit 3

Programming Arduino and Raspberry-PI Introduction to Arduino programming – Arduino GPIO's – Arduino Digital and Analog Input & Output - Interfacing Sensors to Arduino – Interfacing communication devices to Arduino – Configuring Raspberry-PI-Introduction to python-Programming Raspberry-PI using python-Raspberry-PI GPIO's- Interfacing sensors to Raspberry-PI- Communicating Arduino and Raspberry-PI using ethernet / Bluetooth/ Wi-Fi- Remote Actuation and control of motors, LED's and Relays using Arduino and Raspberry-PI.

Unit 4

Introduction to cloud and IOT cloud Services - Cloud services for IOT storage-Introduction to cloud services to visualize IOT data- Streaming IOT data to cloud-Plot and Visualize data using cloud tools- Adding IOT devices to cloud- Integrating Arduino and Raspberry-PI to ThingSpeak /IBM Watson.

Textbooks/References:

The Internet of Things: Enabling Technologies, Platforms, and Use Cases", by Pethuru Raj and Anupama C. Raman (CRC Press).

Singh, R., Gehlot, A., Gupta, L. R., Singh, B., & Swain, M. (2019). Internet of things with Raspberry Pi and Arduino. CRC Press.

Strickland, James R. "Raspberry Pi for Arduino Users." Raspberry Pi for Arduino Users -Building IoT and Network Applications and Devices,(2018).

Singh, Rajesh, Anita Gehlot, Lovi Raj Gupta, Bhupendra Singh, and Mahendra Swain. Internet of things with Raspberry Pi and Arduino. CRC Press, 2019.

Wallace, Shawn, Matt Richardson, and Wolfram Donat. Getting started with raspberry pi. Maker Media, Inc., 2021.

Banzi, Massimo, and Michael Shiloh. Getting started with Arduino. Maker Media, Inc., 2022.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (Minimum 3)	Internal	30
Quizzes (Minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	20

23xxxxxx	Operating Systems	L-T-P-C: 3- 0- 2- 5
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Course Objectives

- To give insight about design and development of Operating Systems
- To introduce the concepts of process creation and synchronization.
- To introduce the memory management techniques used by the Operating System.
- To understand the adaptation of the concepts by modern Operating Systems.

Course Outcomes

After completing this course, students will be able to

CO1	Illustrate the use of system calls to perform basic Operating System functionalities.
CO2	Apply the algorithms for resource management
CO3	Analyze the usage of Synchronization techniques.
CO4	Analyze memory management techniques.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	2	3	2	2	1	-	-	-	2	2	-	1	1	2	1
CO2	2	3	3	2	1	-	-	-	2	2	-	2	1	2	1
CO3	2	2	3	2	1	-	-	-	2	2	-	2	1	2	1
CO4	2	2	3	1	1	-	-	-	2	2	-	2	1	2	1

Syllabus

Unit 1

Operating systems, structure, operating systems services, system calls. Process and Processor management: Process concepts, process scheduling and algorithms, threads, multithreading. CPU scheduling and scheduling algorithms.

Unit 2

Process synchronization, critical sections, Deadlock: Shared resources, resource allocation and scheduling, resource graph models, deadlock detection, deadlock avoidance, deadlock prevention algorithms, mutual exclusion, semaphores, monitors, wait and signal procedures. Memory management: contiguous memory allocation, virtual memory, paging, page table structure, demand paging, page replacement policies, thrashing, segmentation.

Unit 3

Disk scheduling algorithms and policies, File management: file concept, types and structures, directory structure, Case study on Unix (about process management, Thread management and Kernel) and Mobile OS – iOS and Android – Architecture and SDK Framework, Media Layer, Services Layer, Core OS Layer, File System)

Textbook / References

Silberschatz and Galvin, "Operating System Concepts", 10th Edition, Wiley India, 2018.

Tannenbaum A S, "Modern Operating Systems", Prentice Hall India, 2003.

W. Stallings, "Operating Systems: Internals and design Principles", Pearson Ed., LPE, 6th Ed., 2009

X. M.J. Bach, "Design of Unix Operating system", Prentice Hall, 1986

Evaluation Pattern

Assessment	Internal/External	Weightage(%)
Assignments (Minimum 3)	Internal	30
Quiz(Minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term project/End semester examination	External	30

23xxxxxx	Database Management System	L-T-P-C: 2- 0- 2- 4
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Course Objectives

- This course aims to understand the concepts of database design, database languages, database-system implementation and maintenance
- The course will provide knowledge of the design and development of databases for AI applications using SQL and python
- The course will provide an understanding of various databases system including modern databases systems apt for AI and ML applications

Course Outcomes

After completing this course, students will be able to

CO1	Formulate relational algebraic expressions, SQL and PL/SQL statements to query relational databases.
CO2	Build ER models for real world databases.
CO3	Design a normalized database management system for real world databases.
CO4	Apply the principles of transaction processing and concurrency control
CO5	Use high-level right database for AI and ML applications.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	3	3	-	-	-	-	-	-	-	-	-	1
CO2	1	3	3	3	3	-	-	-	-	-	-	-	-	-	1
CO3	2	3	2	3	-	-	-	2	2	2	2	-	-	1	2
CO4	1	1	1	2	-	-	-	-	-	-	-	-	-	-	-
CO5	1	1	-	-	-	-	-	-	-	-	-	-	-	1	2

Syllabus

Unit 1

Introduction: Overview of DBMS fundamentals – Overview of Relational Databases and Keys. Relational Data Model: Structure of relational databases – Database schema – Formal Relational Query Languages – Overview of Relational Algebra and Relational Operations. Database Design: Overview of the design process - The E-R Models – Constraints - Removing Redundant Attributes in Entity Sets - E-R Diagrams - Reduction to

Relational Schemas - Entity Relationship Design Issues - Extended E-R Features – Alternative E-R Notations – Overview of Unified Modelling Language (UML).

Unit 2

Relational Database Design: Features of Good Relational Designs - Atomic Domains and 1NF – Decomposition using Functional Dependencies: 2NF, 3NF, BCNF and Higher Normal Forms. Functional Dependency Theory - Algorithm for Decomposition – Decomposition using multi-valued dependency: 4NF and 4NF decomposition. Database design process and its issues. SQL: review of SQL – Intermediate SQL – Advanced SQL.

Unit 3

Transactions: Transaction concept – A simple transaction model - Storage structure - Transaction atomicity and durability - Transaction isolation – Serializability – Recoverable schedules, Cascade less schedules. concurrency control: Lock-based protocols – Locks, granting of locks, the two-phase locking protocol, implementation of locking, Graph-based protocols. Deadlock handling: Deadlock prevention, Deadlock detection and recovery. Case Study: Different types of high-level databases – MongoDB, Hadoop/Hbase, Redis, IBM Cloudant, Dynamo DB, Cassandra and Couch DB etc. Tips for choosing the right database for the given problem.

Text Books / References

Silberschatz A, Korth HF, Sudharshan S. Database System Concepts. Sixth Edition, TMH publishing company limited; 2017.

Garcia-Molina H, Ullman JD, Widom J. Database System; The complete book. Second Edition, Pearson Education India, 2011

Elmasri R, Navathe SB. Fundamentals of Database Systems. Fifth Edition, Addison Wesley

Evaluation Pattern

Assessment	Internal / External	Weightage (%)
Assignments(minimum 3)	Internal	30
Quiz(minimum 2)	Internal	20
Midterm Examination	Internal	20
Project / End Semester Examination	External	30

23AIE115	User Interface Design	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- Focus in this course is on the basic understanding of user interface design by applying HTML, CSS and Java Script.
- On the completion of the course, students will be able to develop basic web applications .
- This course will serve as the foundation for students to do several projects and other advanced courses in computer science.

Course Outcomes

After completing this course, students will be able to

CO1	Apply the basics of World Wide Web concepts during web development.
CO2	Develop webpage GUI using HTML5 technology.
CO3	Develop GUI using CSS and Java Scrip
CO4	Develop a simple web application using html, CSS and JavaScript.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	2	2	2	-	-	2	2	-	-	-	-	-	-	-	-
CO2	2	2	2	-	-	-	-	-	3	-	-	-	-	-	-
CO3	2	2	2	-	-	-	-	-	3	-	-	-	-	-	-
CO4	-	2	3	3	3	2	2	3	3	3	2	-	-	-	-
CO5	2	2	3	2	3	-	-	3	2	-	-	-	-	-	-

Syllabus

Unit 1

Introduction to Web – Client/Server - Web Server - Application Server- HTML Basics- Tags - Adding Web Links and Images- Creating Tables-Forms - Create a Simple Web Page - HTML 5 Elements - Media – Graphics.

Unit 2

CSS Basics –Features of CSS – Implementation of Borders - Backgrounds- CSS3 - Text Effects -Fonts -Page Layouts with CSS.

Unit 3

Introduction to Java Script –Form Validations – Event Handling – Document Object Model - Deploying an application.

Text Books / References

Kogent Learning Solutions Inc. Html5 Black Book: Covers Css3, Javascript, Xml, Xhtml, Ajax, PhpAndJquery. Second Edition, Dreamtech Press; 2013

Tittel E, Minnick C. Beginning HTML5 and CSS3 For Dummies. Third edition, John Wiley & Sons; 2013.

Powell TA, Schneider F. JavaScript: the complete reference. Paperback edition, Tata McGraw-Hill; 2012.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Strategic Lessons from Mahabharata	L-T-P-C: 2- 0- 2- 3
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Syllabus

Unit 1 – Unit 4

Mahābhārata - A Brief Summary

A Preamble to the Grand Itihāsa

The Unbroken Legacy

Dharmic insights of a butcher

Unit 5 - 8

The Vows we take: Pratijñā

Mahābhārata - The Encyclopaedia for Kingship and Polity Acumen

Karna: The Maestro that Went Wide of the Mark

Strategical Silhouette of An Extraordinary Peace Mission

Unit 9 -11

Yajñaseni, A Woman from Fire.

Popular Regional Tales

Death and Deathlessness

Unit 12 -14

Mahabharata- An All-Encompassing Text

Mahabharatha- Whats and What Nots

Mahābhārata in Adages

Text Books / References

The Mahabharata by Bibek Debroy (Translator)

Some exemplary characters of Mahabharata by Geeta press.

Epic India: by C V Vaidya

The Mahabharata as a History and Drama: by Rai Promatha Nath Mullick

The Mahabharata- A Criticism: by C V Vaidya

The Mahabharata Abridged:by C V Vaidya

23xxxxx	Softskills	L-T-P-C: 2-0-2-4
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Course Outcomes

After completing this course, students will be able to

CO1	Soft Skills: At the end of the course, the students will have the ability to communicate convincingly and negotiate diplomatically while working in a team to arrive at a win-win situation. They would further develop their inter-personal and leadership skills.
CO2	Soft Skills: At the end of the course, the students shall learn to examine the context of a Group Discussion topic and develop new perspectives and ideas through brainstorming and arrive at a consensus
CO3	Aptitude: At the end of the course, students will be able to identify, recall and arrive at

	appropriate strategies to solve questions on geometry. They will be able to investigate, interpret and select suitable methods to solve questions on arithmetic, probability and combinatorics
CO4	Verbal: At the end of the course, the students will have the ability to relate, choose, conclude and determine the usage of right vocabulary
CO5	Verbal: At the end of the course, the students will have the ability to utilise prior knowledge of grammar to recognise structural instabilities and modify them
CO6	Verbal: At the end of the course, the students will have the ability to comprehend, interpret, deduce and logically categorise words, phrases and sentences. They will also have the ability to theorise, discuss, elaborate, criticise and defend their ideas

Syllabus

Professional grooming and practices : Basics of corporate culture, key pillars of business etiquette. Basics of etiquette: Etiquette – socially acceptable ways of behaviour, personal hygiene, professional attire, cultural adaptability. Introductions and greetings: Rules of the handshake, earning respect, business manners. Telephone etiquette: activities during the conversation, conclude the call, to take a message. Body Language: Components, undesirable body language, desirable body language. Adapting to corporate life: Dealing with people.

Group discussions: Advantages of group discussions, structured GD – roles, negative roles to be avoided, personality traits to do well in a GD, initiation techniques, how to perform in a group discussion, summarization techniques.

Listening comprehension advanced: Exercise on improving listening skills, grammar basics: Topics like clauses, punctuation, capitalization, number agreement, pronouns, tenses etc.

Reading comprehension advanced: A course on how to approach middle level reading comprehension passages.

Problem solving level III: Money related problems; Mixtures; Symbol based problems; Clocks and calendars; Simple, linear, quadratic, and polynomial equations; special equations; Inequalities; Functions and graphs; Sequence and series; Set theory; Permutations and combinations; Probability; Statistics.

Data sufficiency: Concepts and problem solving.

Non-verbal reasoning and simple engineering aptitude: Mirror image; Water image; Paper folding; Paper cutting; Grouping of figures; Figure formation and analysis; Completion of incomplete pattern; Figure matrix; Miscellaneous.

Spatial aptitude: Cloth, leather, 2D and 3D objects, coin, match sticks, stubs, chalk, chess board, land and geodesic problems etc., related problems.

Text Books / References

A Communicative Grammar of English: Geoffrey Leech and Jan Svartvik. Longman, London.

Adair. J., (1986), "Effective Team Building: How to make a winning team", London, U.K: Pan Books.

Gulati. S., (2006) "Corporate Soft Skills", New Delhi, India: Rupa & Co.

The Hard Truth about Soft Skills, by Amazone Publication.

Quick Maths – Tyra.

Quicker Arithmetic – Ashish Aggarwal

Test of reasoning for competitive examinations by Thorpe.E.

TMH Non-verbal reasoning by R. S. Aggarwal, S. Chand

Books on GRE by publishers like R. S. Aggrawal, Barrons, Kaplan, The Big Book, and Nova More Games Teams Play, by Leslie Bendaly, McGraw Hill Ryerson.

The BBC and British Council online resources

Owl Purdue University online teaching resources www.thegrammarbook.com.

online teaching resources

www.englishpage.com- online teaching resources and other useful websites.

SEMESTER 5

23xxxxxx	Mathematics for Intelligent Systems 5	L-T-P-C:2-0-2-3
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Course Objectives

- To introduce students to the concepts of probabilistic graphical models and their applications in various fields.
- To teach students the methods of representation learning in Bayesian Networks.
- To enable students to perform inference in Markov Networks and Markov Random Fields.
- To provide an appreciation of probabilistic reasoning required for AI.

Course Outcomes

After completing this course, students will be able to

CO1	Model complex systems using the basics of probabilistic graphical models.
CO2	Develop a mathematical foundation of Bayesian Networks and their applications in real-world scenarios.
CO3	Develop directed and undirected graphical models.
CO4	Apply graphical models to real-world problems such as image recognition, natural language processing, and recommendation systems.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	3	2	3	-	-	-	3	2	-	3	3	-	3
CO2	3	3	3	2	3	-	-	-	3	2	-	3	3	-	3
CO3	3	3	3	2	3	-	-	-	3	2	-	3	3	-	3
CO4	3	3	3	2	3	-	-	-	3	2	-	3	2	-	3

Syllabus

Unit 1

Introduction to probabilistic graphical models, Probabilistic AI, Introduction to Bayesian Networks, Representation Learning in Bayesian Networks, Inference in Bayesian Networks

Unit 2

Markov Networks, Independencies in Markov Networks, Hidden Markov Models

Unit 3

Markov Random Fields (MRF), Decision Networks, From Bayesian Networks to Markov Networks

Text Books / References

Artificial Intelligence: A modern Approach, S J Russell and P Norvig, Pearson (3rd edition), 2010.

Machine Learning: A Probabilistic Perspective, Kevin Murphy and Francis Bach, Penguin Publishers, 2012

Probabilistic graphical models: principles and techniques. Koller, Daphne, and Nir Friedman. MIT press, 2009.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (Minimum 3)	Internal	30
Quiz(Minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term project/End semester examination	External	30

23xxxxxx	Computer Vision	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- To introduce students to the state-of-the-art algorithms in image analysis and object recognition.
- Give an exposure to video analysis techniques for object tracking and motion estimation.
- To build good understanding on the computer vision concepts and techniques to be applied for robotic vision applications.
- Enable students to apply the vision algorithms and develop applications in the domain of image analysis and robotic navigation.

Course Outcomes

After completing this course, students will be able to

CO1	image segmentation.
CO2	object recognition.
CO3	object tracking in videos.
CO4	3D reconstruction

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	2	3	2	3	2	1	2	3	3	2	3	3	1	1
CO2	3	2	3	2	3	2	1	2	3	3	2	3	3	1	1
CO3	3	2	3	2	3	2	1	2	3	3	2	3	3	1	1

CO4	3	2	3	2	3	2	1	2	3	3	2	3	3	1	1
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Syllabus

Unit 1

Image Segmentation Algorithms: contextual, non-contextual segmentation, texture segmentation. Feature Detectors and Descriptors, Feature Matching-Object Recognition, Face detection (Viola Jones), Face Recognition.

Unit 2

Modern computer vision architectures based on deep convolutional neural networks, The Use of Motion in Segmentation Optical Flow & Tracking Algorithms, YOLO, DeepSORT: Deep Learning to Track Custom Objects in a Video, Action classification with convolutional neural networks, RNN, LSTM

Unit 3

Markov Random Fields (MRF), Decision Networks, From Bayesian Networks to Markov Networks Image registration, 2D and 3D feature-based alignment, Pose estimation, Geometric intrinsic calibration, - Camera Models and Calibration: Camera Projection Models - Projective Geometry, transformation of 2-d and 3-d, Internal Parameters, Lens Distortion Models, Calibration Methods Geometry of Multiple views - Stereopsis, Camera and Epipolar Geometry, Fundamental matrix; Homography, Rectification, DLT, RANSAC, 3-D reconstruction framework; Auto-calibration., Introduction to SLAM (Simultaneous Localization and Mapping).

Text Books / References

Ian Goodfellow, Yoshua Bengio, Aaron Courville, Francis Bach, "Deep Learning (Adaptive Computation and Machine Learning series)", January 2017, MIT Press.

Richard Szelinski, "Introduction to Computer Vision and its Application", 2010.

E. Trucco and A. Verri, "Introductory techniques for 3D Computer Vision", Prentice Hall, 1998.

Marco Treiber, "An Introduction to Object Recognition Selected Algorithms for a Wide Variety of Applications", Springer, 2010.

Forsyth and Ponce, "Computer Vision: A Modern Approach", Second Edition, Prentice Hall, 2011.

R. C. Gonzalez, R. E. Woods, 'Digital Image Processing', 4th edition Addison-Wesley, 2016.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (Minimum 2)	Internal	30
Quizzes (Minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

22AIEXXX	Big Data Analytics	L-T-P-C:2-0-2-3
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Course Objectives

- This course aims at introducing the concept of data structure hierarchy.
- It will also expose the students to the basic and higher order data structures.
- Further the students will be motivated to apply the concept of data structures to various engineering problems.

Course Outcomes

After completing this course, students will be able to

CO1	Implement functional and object-oriented programs in Scala, including using higher-order functions, pattern matching, and type classes
CO2	Create and maintain a Spark deployment, including cluster configuration, resource allocation, and job monitoring
CO3	Deploy of Spark for various use cases, such as ETL, data warehousing, and real-time analytics.
CO4	Analyze real-world data sets and extract meaningful insights using statistical and machine learning techniques

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	2	3	-	-	-	3	2	3	3	-	-	-
CO2	3	3	3	3	3	-	-	-	3	2	3	3	-	-	-
CO3	3	2	3	3	3	-	-	-	3	2	3	3	-	-	-
CO4	3	3	3	2	3	-	-	-	3	2	3	3	-	-	-

Syllabus

Unit 1

Introduction to Big Data Analytics: Definition, characteristics, and importance of big data, tools and technologies for big data analytics, State-of-the-art computing paradigms/platforms, Hadoop ecosystem in Brief, Mapper, Reducer.

Unit 2

Introduction to Functional Programming (FP), FP concepts in Scala Programming, Mutable and Immutable Data structures, Scala Collections, Type Hierarchy, Higher Order Functions, Closures, ConsList, Tail Recursion, Object Oriented Programming in Scala, Introduction to concurrency

Unit 3

Basic entity classes and objects in Scala, Spark Architecture, Spark Cluster, Resilient Distributed Datasets (RDDs), Spark Transformations and Actions APIs, DataFrames and Datasets in Spark, Basic Operations on RDDs and DataFrames, lazy evaluations and optimization, Directed Acyclic Graph (DAG)

Unit 4

Introduction to Machine Learning with Spark, MLlib and its algorithms, Building a Machine Learning Pipeline in Spark, Case Study in Healthcare, Finance, etc.

Text Books / References

'Learning Spark: Lightning-Fast Big Data Analysis', Holden Karau , Andy Konwinski, Patrick Wendell and Matei Zaharia, O'Reilly; 1st edition , 2015

'Programming in Scala: A Comprehensive Step-by-Step Guide', Martin Odersky, Lex Spoon and Bill Venners, Artima Inc; Version ed. edition , 2008

'High Performance Spark: Best Practices for Scaling and Optimizing Apache Spark', Holden Karau, Rachel Warren, O'Reilly; 1st edition, 2017

'Scala for the Impatient', Cay S. Horstmann, Addison-Wesley; 2nd edition, 2017

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (Minimum 3)	Internal	30
Quiz(Minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term project/End semester examination	External	30

23xxxxxx	Quantum Computing	L-T-P-C:2-0-2-3
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Course Objectives

- Studying the structural units of quantum computers of the future, forming an understanding of the differences between quantum bits and classical bits.
- Study of basic quantum logical operations and algorithms for processing quantum information.

Course Outcomes

After completing this course, students will be able to

CO1	The basic principles of quantum computing.
CO2	The fundamental differences between conventional computing and quantum computing.
CO3	Several basic quantum computing algorithms.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	3	2	3	-	3	3	3	-	-	-	-	-	-
CO2	3	3	3	2	3	2		-	3	-	-	-	-	-	-
CO3	3	3	3	2	3	-	2	-	3	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Syllabus

Unit 1

Review of the principles of quantum computation, The Deutsch Jozsa Algorithm, Bernstein Vazirani Algorithm, Simmons Algorithm

Unit 2

Shor's Algorithm, Grover's Search, Grover's algorithm Programming

Unit 3

NISQ-era quantum algorithms, Variational Quantum Algorithms

Unit 4

Variational Quantum Eigen solver, Quantum Generative Adversarial Networks (QGANs)

Unit5.

Applications

Text Books / References

Bernhardt, Chris. Quantum computing for everyone. Mit Press, 2019.

Yuly Billing, Quantum Computing for High School Students.

Quantum Computation and Quantum Information, M. A. Nielsen & I.Chuang, Cambridge University Press (2000).

Gilbert Strang, Introduction to Linear Algebra, Fifth Edition, Wellesley-Cambridge Press, 2016.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (Minimum 3)	Internal	30
Quiz(Minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term project/End semester examination	External	30

23xxxxxx	High-Performance and Cloud Computing	L-T-P-C: 2- 0 -2- 3
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Course Objectives

- Familiarize student with architectural overview of modern HPC and GPU based heterogeneous architectures, focusing on its computing power versus data movement needs.
- Familiarize the students working with cloud platforms and services to configure and use computational resources and storage.
- To educate students how to write efficient parallel programming and GPU programming.
- To discuss various application of HPC computational techniques in computational science.

Course Outcomes

After completing this course, students will be able to

CO1	Apply high-performance computing in different research field.
CO2	Design OpenMPI programme and CUDA programme
CO3	Simulate on cloud computing system.
CO4	Evaluate how the convergence of HPC and AI is transforming the data science.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	2	-	2	2	3	2	-	-	2	1	3	1	1	-
CO2	2	3	3	3	3	-	1	-	2	1	2	3	3	3	1

CO3	2	2	3	2	3	1	1	-	2	1	2	3	2	2	1
CO4	2	2	1	2	3	2	1	1	2	1	2	3	3	3	3

Syllabus

Unit 1

Introduction to basic architecture and OS concepts –architecture of parallel computing–shared and distribution memory in parallel computing – parallel algorithm – performance metrics of parallel algorithm.

Unit 2

Introduction to OpenMP – essentials of OpenMP – data sharing and synchronization – efficient OpenMP for matrix computing – Introduction to MPI and distributed memory parallel computing – communicating using MPI – Matrix representation of physical system and parallel matrix solvers – domain decomposition techniques

Unit 3

Overview of GPU architecture and its evolution –introduction to GPGPU and CUDA – CUDA programming – thread execution in CUDA programming – matrix computing in CUDA -introduction to cuBLAS and cuDNN libraries for linear algebra and deep learning - case studies of GPU: accelerated applications in scientific computing, data analytics, and machine learning

Unit 4

Introduction to cloud computing and its importance – benefits and challenges of cloud computing - types of cloud services (IaaS, PaaS, SaaS) and their characteristics - cloud computing architecture and its components - cloud storage and its types - cloud networking and its challenges - cloud security and its importance - cloud application - benefits and challenges of HPC and AI - synergy between HPC and AI - training and inference of AI models using HPC

Text Books / References

"High Performance Computing: Modern Systems and Practices" by Thomas Sterling and Matthew Anderson

"CUDA by Example: An Introduction to General-Purpose GPU Programming" by Jason Sanders and Edward Kandrot

"Parallel Programming with MPI" by Peter S. Pacheco

Comer, D. (2021). The Cloud Computing Book: The Future of Computing Explained. CRC Press

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (Minimum 2)	Internal	20
Quiz(Minimum 2)	Internal	20
Mini Project	Internal	20
Mid-Term Examination	Internal	10
Term project/End semester examination	External	30

23xxxxxx

Control Systems

L-T-P-C: 2- 0- 2- 3

Course Objectives

- This course introduces the fundamentals of control systems through a hands-on approach involving programming tools such as MATLAB.
- This course familiarizes concepts of control systems, such as open-loop, closed-loop, and feedback systems.
- This course enables the students to judge the performance and stability of control systems

Course Outcomes

After completing this course, students will be able to

CO1	Explain the fundamental principles that govern control systems.
CO2	Apply analytical techniques to evaluate and characterize basic control systems.
CO3	Evaluate the performance and stability of control systems
CO4	Apply control system theory to practical applications in engineering.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	1	3	-	-	-	3	3	-	2	3	2	1
CO2	3	3	3	2	3	1	-	-	3	3	-	2	2	2	1
CO3	3	3	3	2	3	-	-	-	3	3	-	2	2	2	1
CO4	3	3	3	2	3	1	-	-	3	3	-	3	3	2	1

Syllabus

Unit 1

Introduction to Control Systems and Frequency Domain Methods, Definition and types of control systems, Mathematical modelling of systems, Block diagram and signal flow graph representation of systems, Feedback and control system characteristics, Stability analysis and root locus techniques, Frequency response analysis, Bode plot and Nyquist plot, PID controllers Lead-lag compensators, Design of classical controllers using root locus.

Unit 2

State Space Methods State space analysis and design, State-Space representation of control systems: state variables, state-space models, Multivariable control systems: MIMO systems, decoupling, Controllability and observability, Pole placement and observer design, Linear quadratic regulator (LQR) Optimal control, Introduction to nonlinear control

Unit 3

Applications of Control Systems, Control of mechanical systems, Control of electrical systems, Control of chemical and biological systems, Introduction to optimal control for aerospace system

Text Books / References

S. L. Brunton and J. N. Kutz, "Data-driven science and engineering: Machine learning, dynamical systems, and control", Cambridge University Press, 2022. ISBN 9781108422093.

Ogata Katsuhiko, "Modern control engineering", Prentice Hall, 2010. ISBN 9780136156734.

M. F. Golnaraghi, B. C. Kuo, and M. F. Golnaraghi, "Automatic control systems" Wiley, 2010. ISBN9780470048962.

N. Nise, “Control systems engineering”, 6th ed. John Wiley & Sons, 2017. ISBN 9780470917695.

G. F. Franklin, J. D. Powell, and M. L. Workman, “Digital control of dynamic systems”, Vol. 3, Ellis Kagle Press, 1998. ISBN 9780979122606.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (Minimum 3)	Internal	30
Quiz(Minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term project/End semester examination	External	30

SEMESTER 6

23xxxxxx

Mathematics for Intelligent Systems 6

L-T-P-C:2-0-2-3

Course Objectives

- To introduce students to the concept of Neuro-symbolic AI and its significance in artificial intelligence.
- To provide an overview of knowledge graphs and their applications in various domains

Course Outcomes

After completing this course, students will be able to

CO1	Develop intelligent systems using the concept of Neuro-Symbolic AI.
CO2	Develop knowledge representation and reasoning techniques in Neuro-Symbolic AI.
CO3	Apply the concepts of logical neural networks and Markov random fields in Neuro-Symbolic AI
CO4	Develop hybrid models that combine different AI approaches, such as Neuro-Symbolic AI and deep learning

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	3	2	3	-	-	-	3	2	-	3	3	-	3
CO2	3	3	3	2	3	-	-	-	3	2	-	3	3	-	3
CO3	3	3	3	2	3	-	-	-	3	2	-	3	3	-	3
CO4	3	3	3	2	3	-	-	-	3	2	-	3	2	-	3

Syllabus

Unit 1

Introduction to Neuro-Symbolic AI: Definition and overview of Neuro-Symbolic AI- Advantages and disadvantages of Neuro-Symbolic AI- Applications of Neuro-Symbolic AI.

Unit 2

Knowledge Representation and Reasoning: Reasoning in neuro-symbolic AI - Types of reasoning. Logical Neural Networks-Markov Random Fields-Hybrid Models

Unit 3

Explainable AI, Multi-Modal Neuro-Symbolic AI, Future Directions in Neuro-Symbolic AI

Text Books / References

Bouneffouf, Djallel, and Charu C. Aggarwal. "Survey on Applications of Neurosymbolic Artificial Intelligence." *arXiv preprint arXiv:2209.12618* (2022).

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Analog Computing	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- Learn the principles of analog computing.
- Learn the concepts of Hybrid computing.
- Apply machine learning in Hybrid computing.

Course Outcomes

After completing this course, students will be able to

CO1	Understand the principles of analog computing
CO2	Understand the concepts of Hybrid computing (Analog and digital computing
CO3	Computation using Hybrid computing
CO4	Apply machine learning in Hybrid computing

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	-	-	-	-	-	3	3	-	-	-	-	2	-	-	-
CO2	-	-	-	-	-	1	-	3	-	-	-	2	-	-	-
CO3	-	-	-	-	-	3	3	3	-	-	-	2	-	-	-
CO4	-	-	-	-	-	3	3	3	-	-	-	2	-	-	-

Syllabus

Unit 1

Introduction to analog computer, direct versus indirect analogies, A short history of analog computing, Characteristics of analog computers, Computing elements – machine units, summer, integrators, free elements, potentiometers, function generators, multiplication, comparators and switches, Input/Output devices.

Unit 2

Analog computer operation, basic programming – radioactive decay, analytical solutions, using an analog computer, scaling, harmonic functions, sweep, Mathematical pendulum, Straight forward implementation, variants, Mass-spring-damper system, analytical solution, using an analog computer, RLC-circuit

Unit 3

Inverse functions, square root, division, power and polynomials, low pass filter, triangle/square wave generator, Ideal diode, absolute value, limiters, dead space, hysteresis, Bang-bang, Minimum/Maximum holding circuits, sample and hold, time derivative, time delay, historic approaches to delay, digitization, sample and hold circuits, analog delay network.

Unit 4

Inverse Chemical kinetics, damped pendulum with external force, MATHIEU'S equation, Introduction, Scaling and programming, VANDER POL's equation, Programming, Solving the one dimensional SCHRODINGER equation, Ballistic trajectory, Charged particle in a magnetic field, RUTHERFORD scattering, Celestial mechanics, Bouncing ball, Zombie apocalypse, ROSSLER attractor, LORENZ attractor, another Lorenz attractor, CHAU attractor, Nonlinear chaos, AIZAWA attractor, NOSE-HOOVER oscillator, rotating spiral, flow around an airfoil, Heat transfer, Two dimensional heat transfer, systems of linear equations, Human-in-the-loop, Inverted pendulum, Double pendulum

Unit 5

Hybrid computing - Hybrid controllers, Basic operations, shell trajectory, data gathering, training an AI with analog computer.

Textbooks / References

Analog and Hybrid computer programming, Bernd Ulmann, CPI books GmbH, Leck

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Reinforcement Learning	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- This course will provide a solid introduction to the field of reinforcement learning.
- It will also make the students learn about the core challenges and approaches, including exploration and exploitation.
- The course will make the students well versed in the key ideas and techniques for reinforcement learning.

Course Outcomes

After completing this course, students will be able to

CO1	Formulate an application problem as a reinforcement learning problem
CO2	Implement common reinforcement learning algorithms using Python/MATLAB
CO3	Evaluate reinforcement learning algorithms on the metrics such as regret, sample complexity,

	computational complexity, empirical performance, and convergence
CO4	Evaluate different approaches for addressing exploration vs exploitation challenge in terms of performance, scalability, complexity of implementation, and theoretical guarantees

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	2	2	2	2	2	-	-	1	-	-	-	2	3	-	1
CO2	2	2	2	2	2	-	-	1	-	-	-	2	3	-	1
CO3	1	1	1	1	1	-	-	1	-	-	-	1	-	3	1
CO4	1	1	1	1	1	-	-	1	-	-	-	1	-	3	1

Syllabus

Unit 1

Introduction to Reinforcement Learning – History of Reinforcement Learning - Elements of Reinforcement Learning – Limitations and scope

Unit 2

Multi-armed Bandits – Finite Markov Decision Processes – Dynamic Programming – Policy evaluation – Policy improvement – Policy Iteration – Value Iteration

Unit 3

Monte Carlo Methods – Monte Carlo prediction – Monte Carlo control – Incremental Implementation – Temporal-Difference Learning – TD prediction – Q-Learning - n-step Bootstrapping

Unit 4

Planning and Learning with Tabular Methods – Models and planning – Prioritized sweeping – Trajectory sampling – Heuristic search – Rollout algorithms

Text Books / References

Richard.S.Sutton and Andrew G.Barto, Reinforcement Learning, MIT Press, Second Edition, 2018

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

Syllabus**Unit 1 – Unit 4**

Educational Heritage of Ancient India

Life and Happiness

Impact of Colonialism and Decolonization

A timeline of Early Indian Subcontinent

Unit- 5 - 8

Pinnacle of Selflessness and ultimate freedom

Indian approach towards life

Circle of Life

Ocean of love; Indian Mahatmas.

Unit 9 - 13

Man's association with Nature

Celebrating life 24/7.

Metaphors and Tropes

Become A Strategic Thinker (Games / Indic activity)

India: In the Views of Other Scholars and Travellers

Unit 14-16

Personality Development Through Yoga.

Hallmark of Indian Traditions: Advaita Vedanta, Theory of oneness

Conversations on Compassion with Amma

Text Books / References

- *The beautiful tree* by Dharampal
- *Peasants and Monks in British India* by William Pinch
- *India, that is Bharat: Coloniality, Civilisation, Constitution* by J Sai Deepak
- *Awaken Children Dialogues with Mata Amritanandamayi*
- *Man and Nature* by Mata Amritanandamayi Devi
- *What Becomes of the Soul After Death*, Divine Life Society

SEMESTER 7

23xxxxxx	ENVIRONMENTAL SCIENCE	P/F
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Course Objectives

- To study the nature and facts about environment
- To appreciate the importance of environment by assessing its impact on the human world
- To study the integrated themes and biodiversity, pollution control and waste management

Course Outcomes

After completing this course, students will be able to

CO1	Ability to understand aspects of nature and environment
CO2	Ability to analyse impact of environment on human world.
CO3	Ability to comprehend pollution control and waste management

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PS O3
CO															
CO1	-	-	-	-	-	3	2	3	-	-	-	-	-	-	-
CO2	-	-	-	-	-	3	2	3	-	-	-	-	-	-	-
CO3	-	-	-	-	-	3	2	3	-	-	-	-	-	-	-

Syllabus

Unit 1

Over view of the global environment crisis – Biogeochemical cycles – Climate change and related international conventions and treaties and regulations – Ozone hole and related International conventions and treaties and regulations – Overpopulation – energy crisis – Water crisis – ground water hydrogeology – surface water resource development.

Unit 2

Ecology, biodiversity loss and related international conventions – treaties and regulations – Deforestation and land degradation – food crisis – water pollution and related International and local conventions – treaties and regulations – Sewage domestic and industrial and effluent treatment – air pollution and related international and local conventions – treaties and regulations – Other pollution (land, thermal, noise).

Unit 3

Solid waste management (municipal, medical, e-waste, nuclear, household hazardous wastes) – environmental management – environmental accounting – green business – eco-labelling – environmental impact assessment – Constitutional – legal and regulatory provisions – sustainable development, Topic modelling

Text Books / References

R. Rajagopalan, "Environmental Studies – From Crisis to Cure", Oxford University Press, 2005, ISBN 0-19-567393-X.

G.T.Miller Jr., "Environmental Science", 11th Edition, Cenage Learning Pvt. Ltd., 2008.

Benny Joseph, "Environmental Studies", Tata McGraw-Hill Publishing company Limited, 2008.

Evaluation Pattern

Assessment	Internal	External
Online Test	-	100
		P/F

23xxxxxx	INDIAN CONSTITUTION	L-T-P-C: P/F
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Course Objectives

- To know about Indian constitution.
- To know about central and state government functionalities in India
- To know about Indian society

Course Outcomes

After completing this course, students will be able to

CO1	Understand the functions of the Indian government
CO2	Understand and abide the rules of the Indian constitution.
CO3	Understand and appreciate different culture among the people

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	-	-	-	-	-	3	2	3	-	-	-	-	-	-	-
CO2	-	-	-	-	-	3	2	3	-	-	-	-	-	-	-
CO3	-	-	-	-	-	3	2	3	-	-	-	-	-	-	-

Syllabus

Unit 1

Historical Background – Constituent Assembly Of India – Philosophical Foundations Of The Indian Constitution– Preamble – Fundamental Rights – Directive Principles Of State Policy – Fundamental Duties – Citizenship – Constitutional Remedies For Citizens

Unit 2

Union Government – Structures of the Union Government and Functions – President – Vice President – Prime Minister – Cabinet – Parliament – Supreme Court of India – Judicial Review.

Unit 3

State Government – Structure and Functions – Governor – Chief Minister – Cabinet – State Legislature – Judicial System in States – High Courts and other Subordinate Courts.

Text Books

Durga Das Basu, “Introduction to the Constitution of India “, Prentice Hall of India, New Delhi.

R.C.Agarwal, (1997) “Indian Political System”, S.Chand and Company, New Delhi.

References

Sharma, Brij Kishore, “Introduction to the Constitution of India”, Prentice Hall of India, New Delhi..

Evaluation Pattern

Assessment	Internal	External
Online Test	-	100
		P/F

23xxxxxx	PROJECT PHASE – 1	L-T-P-C: 0- 0- 18- 6
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Course Objectives

- Project Phase – 1 aims at helping students to identify the research problems by conducting a thorough
- literature review
- The course introduces the students to real world problems associated with AI
- The course also aims at helping students to publish scientific articles in peer reviewed scientific
- publications.

Course Outcomes

After completing this course, students will be able to

CO1	Identify a valid research problem by conducting literature review in the appropriate area
CO2	Identify the appropriate methodology to solve the research problem..
CO3	Apply the AI tools & techniques to solve the identified problem.
CO4	Communicate scientific discoveries through peer-reviewed publications.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	3	3	3	2	2	2	3	3	3	3	2	3	-
CO2	3	3	3	3	3	2	2	2	3	3	3	3	2	3	-
CO3	3	3	3	3	3	2	2	2	3	3	3	3	2	3	-

CO4	3	3	3	3	3	2	2	2	3	3	3	3	-	-	3
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Evaluation Pattern

Assessment	Weightage (%)
Internal	70
External	30

SEMESTER 8

23xxxxxx

PROJECT PHASE – 2

L-T-P-C: 0- 0- 30- 10

Course Objectives

- Project Phase – 2 aims at helping students to solve the identified research problem
- The course introduces the students to real world problems associated with AI
- The course also aims at helping students to publish scientific articles in peer reviewed scientific publications.

Course Outcomes

After completing this course, students will be able to

CO1	Solve a valid research problem by employing appropriate tools & techniques.
CO2	Implement the appropriate methodology to solve the research problem.
CO3	Apply the AI tools & techniques to solve the identified problem.
CO4	Communicate scientific discoveries through peer-reviewed publications.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	3	3	3	2	2	2	3	3	3	3	2	3	-
CO2	3	3	3	3	3	2	2	2	3	3	3	3	2	3	-
CO3	3	3	3	3	3	2	2	2	3	3	3	3	2	3	-
CO4	3	3	3	3	3	2	2	2	3	3	3	3	-	-	3

Evaluation Pattern

Assessment	Weightage (%)
Internal	70
External	30

23xxxxxx

PROJECT PHASE – 2

L-T-P-C: 0- 0- 30- 10

Course Objectives

- Project Phase – 2 aims at helping students to solve the identified research problem.
- The course introduces the students to real world problems associated with AI.
- The course also aims at helping students to publish scientific articles in peer reviewed scientific publications.

Course Outcomes

After completing this course, students will be able to

CO1	Solve a valid research problem by employing appropriate tools & techniques.
CO2	Implement the appropriate methodology to solve the research problem.
CO3	Apply the AI tools & techniques to solve the identified problem.
CO4	Communicate scientific discoveries through peer-reviewed publications.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	3	3	3	2	2	2	3	3	3	3	2	3	-
CO2	3	3	3	3	3	2	2	2	3	3	3	3	2	3	-
CO3	3	3	3	3	3	2	2	2	3	3	3	3	2	3	-
CO4	3	3	3	3	3	2	2	2	3	3	3	3	-	-	3

Evaluation Pattern

Assessment	Weightage (%)
Internal	70
External	30

ELECTIVES

xxxxxxx	Introduction to Robotics	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- This course aims to provide students with a solid foundation in the principles of robot kinematics, dynamics, control, navigation, and localization for both industrial and mobile robots.
- This course intends to equip students with the ability to design and analyze common robotic systems using mathematical and computational tools.
- This course aims to give students hands-on experience in programming robots for navigation and control, enabling them to apply their knowledge to practical applications in the field of robotics.

Course Outcomes

After completing this course, students will be able to

CO1	Develop mathematical models for robot kinematics, dynamics, and control systems to solve problems related to robotic systems.
CO2	Implement simple robotic control systems using different types of control algorithms, such as PD, PID, and adaptive control.
CO3	Develop machine learning/deep learning models for solving NLP applications Apply gained knowledge of robotic concepts to design and program robots for tasks such as path/trajectory planning, localization, and obstacle avoidance.
CO4	Use simulation tools to model and analyze the behavior of robotic systems.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	1	2	1	1		3	3		2	2	-	-
CO2	3	3	1	1	2	1	1		3	3		2	2	1	-
CO3	3	3	2	1	2	1	1	1	3	3		2	3	2	1
CO4	3	3	3	3	3	1	1	1	3	3		3	1	1	1

Syllabus

Unit 1: Industrial Robots – Kinematics

Introduction to Industrial Robots and their Applications, Robot Coordinates and Transformations, Forward Kinematics, DH parameters and table, Twists, Inverse Kinematics, Differential kinematics, Trajectory Planning

Unit 2: Industrial Robots - Dynamics and Control

Robot Dynamics, Euler-Lagrange's Equations, Newton-Euler method, Inverse Dynamics and Trajectory Tracking, PD & PID Control, Adaptive Control, Force Control.

Unit 3: Mobile Robots - Kinematics, Navigation, and Localization

Introduction to Mobile Robots and their Applications, Mobile Robot Kinematics, Robot Navigation and Path Planning, Localization Techniques, Simultaneous Localization and Mapping (SLAM)

Text Books / References

Corke P. *Robotics, Vision and Control*. Springer; 2017.

Craig J J. *Introduction to Robotics: Mechanics and Control*. Pearson Publishing, 2017.

Spong M.W., Hutchinson S. and Vidyasagar M, *Robot Modeling and Control*, Wiley, 2006

Saha S K, *Introduction to Robotics*, McGraw Hill publishing, 2014

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

xxxxxxxxx	Introduction to ROS2	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- This course aims to introduce students to the fundamentals of Robotic Operating System (ROS2) and Gazebo simulation, including key concepts such as nodes, topics, services, and actions, as well as the development of ROS2 packages.
- This course aims to equip students with the knowledge and skills required to apply ROS2 to mobile and industrial robots, including navigation, SLAM, robot arm control, perception, and communication.
- This course aims to provide students with hands-on experience working with ROS2 and Gazebo, allowing them to design, implement, and test robotic applications using these tools.

Course Outcomes

After completing this course, students will be able to

CO1	Program ROS2 packages and utilize ROS2 concepts such as nodes, topics, services, and actions to control, perceive, and communicate with mobile and industrial robots.														
CO2	Apply ROS2 to mobile robots to navigate, use SLAM, and control them effectively.														
CO3	Apply ROS2 to industrial robots to control robotic arms and perform real world tasks within a simulation environment.														
CO4	Evaluate the performance of chatbots using various metrics and techniques. Use Gazebo to simulate robotic applications and test their code effectively, improving the overall proficiency with ROS2.														

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	1	1	3	1	1	2	1	1	1	1		2	1		
CO2	3	3	3	2	3	1		1	3	3		2	2	2	2
CO3	3	3	3	2	3	1		1	3	3		2	2	2	2
CO4	1	2	2	2	3	-			3	3		2	2	2	1

Syllabus

Unit 1: Introduction to ROS2

Overview of ROS2 architecture and communication protocols, Comparison with ROS1, setting up ROS2 environment and creating a ROS2 package, Introduction to ROS2 command-line tools, Basic ROS2 concepts such as nodes, topics, messages, and services, Introduction to Gazebo simulator.

Unit 2: Advanced ROS2 Concepts

ROS2 middleware and communication mechanisms, ROS2 launch files and parameter management, ROS2 package dependencies and ROS2 ecosystem, Advanced ROS2 concepts such as actions and transformations.

Unit 3: ROS2 Applications in Mobile and Industrial Robotics

Introduction to ROS2-based mobile robot navigation, Overview of ROS2-based industrial robot control, Integration of ROS2 with sensors and actuators for mobile and industrial robots, ROS2-based robot perception and manipulation.

Text Books / References

Rico, F. M. A Concise Introduction to Robot Programming with ROS2. CRC Press; 2023.

Nehmzow U. Mobile robotics: a practical introduction. Springer Science & Business Media; 2012.

P. Corke, Robotics, Vision, and Control, Springer, 2011.

Ros2 Humble Tutorial: <https://docs.ros.org/en/humble/Tutorials.html>

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

xxxxxxx	NLP for Robotics	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- The course aims to introduce spoken language technology with an emphasis on dialog and conversational systems
- The course helps in establishing the understanding of Deep learning and other methods for automatic speech recognition, speech synthesis systems for robotics

Course Outcomes

After completing this course, students will be able to

CO1	Demonstrate understanding of acoustic phonetics in the context of spoken language.
CO2	Analyze different types of dialog systems and their applications.
CO3	Apply AI techniques used in dialog systems.
CO4	Implement automatic speech recognition, text-to-speech synthesis, and evaluation.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	3	2	2	1	1	2	2	1	3	2	0	2
CO2	3	3	3	3	2	2	1	2	2	3	2	3	0	0	0
CO3	3	3	3	3	3	2	1	2	3	2	2	3	2	2	0
CO4	3	3	3	3	3	2	1	2	3	3	2	3	2	2	1

Syllabus

Unit 1

Introduction and Acoustic Phonetics, Overview of dialog: Human conversation. Task-oriented dialog. Dialog systems, Machine Learning in Dialog- Recurrent NNs, Attention, Transformers

Unit 2

Automatic Speech Recognition, Foundation models for spoken language-Using the Speech Brain ASR toolkit, Advanced ASR

Unit 3

Text to Speech (TTS): Overview. Text normalization, Spectrogram prediction, Vocoding, TTS, Evaluation.

Text Books / References

Dan Jurafsky and James H. Martin. Speech and Language Processing, (3rd ed. draft), available at <https://web.stanford.edu/~jurafsky/slp3/>

Yoav Goldberg. A Primer on Neural Network Models for Natural Language Processing. Available at <https://u.cs.biu.ac.il/~yogo/nnlp.pdf>

Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. Available at <https://www.deeplearningbook.org/>

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

xxxxxxx

Robotics Vision

L-T-P-C: 2- 0- 2- 3

Course Objectives

- This course introduces the geometry of image formation and its use for 3D reconstruction and calibration.
- This course introduces the analysis of patterns in visual images that are used to reconstruct and understand objects and scenes.

Course Outcomes

After completing this course, students will be able to

CO1	Apply image formation and camera calibration for various applications.
CO2	Analyze and select image features and apply for image matching.
CO3	Develop image recognition algorithms
CO4	Develop stereo vision applications for distance estimation.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	-	2	2	1	1	1	-	-	-	-	-	-
CO2	3	3	2	3	3	3	2	1	2	1	-	-	-	-	-
CO3	3	3	3	3	3	3	2	3	3	2	-	-	-	-	-
CO4	3	3	1	2	3	2	1	1	1	1	-	-	-	-	-

Syllabus

Unit 1

Introduction, Image Formation – geometric primitives and transformations, photometric image formation, digital camera, Camera calibration. Edge Detection, Segmentation.

Unit 2

Feature Detection and Matching – points and patches, edges, lines, Feature-Based Alignment - 2D, 3D feature-based alignment, pose estimation, Image Stitching, Dense motion estimation – Optical flow - layered motion, parametric motion, Structure from Motion.

Unit 3

Recognition – object detection, face recognition, instance recognition, category recognition, Stereo

Correspondence – Epipolar geometry, 3D reconstruction.

Text Books / References

Szeliski R. Computer Vision: Algorithms and Applications Springer. New York. 2010..

Shapiro LG, Stockman GC. Computer Vision: Theory and Applications. 2001.

Forsyth DA, Ponce J. Computer Vision: a modern approach;2012.

Davies ER. Machine vision: theory, algorithms, practicalities. Elsevier; 2004 Dec 22.

Jain R, Kasturi R, Schunck BG. Machine vision. New York: McGraw-Hill; 1995 Mar 1

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
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Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

xxxxxxx	Mobile Robotics	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- This course aims to provide students with a comprehensive understanding of the underlying principles and technologies used in designing, implementing, and controlling autonomous mobile robots.
- This course seeks to equip students with practical skills in perception and control of mobile robots, including sensing and estimation, basic and advanced control techniques, and sensor fusion and localization.
- This course intends to introduce students to advanced topics in mobile robotics, such as path planning and navigation, motion planning and control, multi-robot systems, and human-robot interaction, providing them with the knowledge necessary to develop and implement cutting-edge mobile robotic applications.

Course Outcomes

After completing this course, students will be able to

CO1	Design autonomous mobile robot systems using the fundamental principles and technologies.
CO2	Apply perception and control techniques to mobile robotics, to solve real-world mobile robotic challenges.
CO3	Use advanced techniques in mobile robotics to design mobile robotic systems that can operate in complex and dynamic environments.
CO4	Analyze the capabilities and limitations of mobile robotic systems in the light of potential impact on society and various industries.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	3	2	3	2	2	2	3	3	-	2	2	1	-
CO2	3	3	3	2	3	2	2	2	3	3	-	2	2	2	1
CO3	3	3	3	3	3	2	2	2	3	3	-	2	2	2	2
CO4	3	3	3	3	3	3	2	3	3	3	-	2	2	2	3

Syllabus

Unit 1: Introduction to Mobile Robotics

Introduction to Mobile Robotics, Brief history of Mobile Robotics, Overview of Mobile Robotics applications, Locomotion Systems in Mobile Robotics, Sensors in Mobile Robotics.

Unit 2: Perception and Control

Perception: Sensing and Estimation, Control: Basic and Advanced Techniques, Sensor Fusion and Localization, Mapping and SLAM (Simultaneous Localization and Mapping).

Unit 3: Advanced Topics in Mobile Robotics

Path Planning and Navigation, Motion Planning and Control, Multi-robot Systems, Human-Robot Interaction.

Text Books / References

Siegwart R, Nourbakhsh IR, Scaramuzza D. Introduction to autonomous mobile robots. MIT press; 2011.

Nehmzow U. Mobile robotics: a practical introduction. Springer Science & Business Media; 2012.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

xxxxxxxx	Artificial Intelligence for Robotics	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- This course aims to introduce students to the fundamentals of artificial intelligence and machine learning techniques and their applications to robotics, including perception, planning, and control.
- This course will help students understand the challenges involved in applying AI and machine learning techniques to robotics and develop the ability to design and implement intelligent robotics systems.
- This course will provide students with the knowledge and skills required to apply deep learning algorithms to robotics problems, including object recognition, motion planning, and robot control.

Course Outcomes

After completing this course, students will be able to

CO1	Apply machine learning and deep learning algorithms to solve robotics problems.
CO2	Design intelligent agents that can perceive and act in different environments.
CO3	Evaluate different robotics and AI applications and identify their strengths and weaknesses.
CO4	Implement deep learning models for robotics tasks.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	1	3	-	-	2	3	2	-	2	3	3	2
CO2	3	3	1	1	3	1	-	-	3	3	-	2	3	3	3
CO3	3	3	3	2	3	2	2	2	3	3	-	2	3	3	3
CO4	3	3	3	2	3	-	-	-	3	3	-	2	3	3	2

Syllabus

Unit 1: Introduction to AI for Robotics

Overview of AI for robotics, Robotics components and their interactions, Sensing and perception in robotics, Planning and decision-making in robotics, Control systems and robotics.

Unit 2: Machine Learning for Robotics

Supervised learning and its applications in robotics, Unsupervised learning and its applications in robotics, Reinforcement learning and its applications in robotics, Transfer learning in robotics

Unit 3: Deep Learning for Robotics

Neural networks and deep learning, Convolutional Neural Networks for Perception in Robotics, Recurrent Neural Networks for Robotics Planning, Deep Reinforcement Learning for Robotics Control

Text Books / References

Govers, F. X. Artificial Intelligence for Robotics. Packt Publishing; 2018

Nehmzow U. Mobile robotics: a practical introduction. Springer Science & Business Media; 2012.

P. Corke, Robotics, Vision, and Control, Springer, 2011.

P. Antsaklis and K. Passino, An Introduction to Intelligent and Autonomous Control, Kluwer, 1993.

D. Kortenkamp, R. Bonasso, and R. Murphy, ed., Artificial Intelligence and Mobile Robots, AAAI Press, 1998.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

xxxxxxx	Introduction to Biomedical Informatics	L-T-P-C: 2- 0 -2- 3
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Course Objectives

- The course is aimed at educating students on the basic concepts and techniques of AI
- This would cover the methods of collecting, cleaning, and preparing biomedical data before processing.
- Introduced different methods for feature set optimization and handling imbalanced biomedical data.
- Describe the role of biomedical informatics in clinical decision-making and the design of clinical decision support systems.

Course Outcomes

After completing this course, students will be able to

CO1	Application of AI and ML to biomedical problems.
CO2	Application data acquisition, integration, retrieval, and manipulation.
CO3	Develop knowledge on different biomedical area, like public health informatics, clinical trials, agro informatics, nutri informatics etc.
CO4	Develop skill to initiate collaborative research with interdisciplinary teams to develop innovative solutions to real-world problems in biomedical informatics using AI techniques and tools.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	2	3	2	3	3	2	-	3	2	-	3	3	-	-
CO2	3	2	3	2	3	3	2	-	3	2	-	3	-	3	2
CO3	3	2	3	2	3	3	2	-	3	2	-	3	-	3	2
CO4	3	2	3	2	3	3	2	-	3	2	-	3	-	2	3

Syllabus

Unit 1

Introduction to Biomedical Informatics – different biomedical data and databases – methods of data integration of heterogeneous data – data retrieval and acquisition - Querying biomedical databases using SQL – heterogeneous data standardization.

Unit 2

Application of data mining and AI techniques in biomedical information – Querying biomedical databases using SQL- Biomedical Imaging Informatics.

Unit 3

Clinical Informatics and Electronic Health Record Systems – AI and mathematical model in clinical trial – community and population health - tackles disease treatment and prevention.

Unit 4

Agro informatics - nutri informatics - ethics of AI in biomedical informatics

Text Books / References

Biomedical Informatics: Computer Applications in Health Care and Biomedicine" by Edward H. Shortliffe and James J. Cimino

"Statistical Methods for Clinical Trials" by J. Rick Turner

"Agro-Informatics and Precision Agriculture" by Manoj Karkee, Qin Zhang, and Shrinivasa K. Upadhyaya

"Public Health Informatics and Information Systems" by J.A. Magnuson and P.G. Fu

Johnson, K. B., & Lorenzi, N. M. (2017). Health Informatics: An Interprofessional Approach. Mosby.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20

Term Project/ End Semester Examination	External	30
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xxxxxxx	Transcriptomic, Proteomics and Metabolomics	L-T-P-C: 2- 0 -2- 3
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Course Objectives

- Introduce students to using artificial intelligence (AI) in transcriptomics, proteomics, and metabolomics research, including machine learning algorithms, data mining techniques, and other computational approaches.
- Teach students how to apply AI techniques to analyse and interpret omics data, including genomic, transcriptomic, proteomic, and metabolomic data, and integrate data from multiple omics platforms.
- Foster students' ability to critically evaluate the strengths and limitations of AI-based approaches in omics research and understand the importance of validation and reproducibility in scientific research.

Course Outcomes

After completing this course, students will be able to

CO1	Application of AI in omics research.
CO2	Develop the ability to integrate data from multiple omics platforms using AI-based approaches.
CO3	Identify limitations of existing AI-based software tools and overcome the computational limitations of models
CO4	Develop critical thinking and problem-solving skills (real-world problems in biomedical, agriculture etc.) to design and carry out experiments in transcriptomics, proteomics, and metabolomics and analyse and interpret the resulting data.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	1	-	-	2	2	-	1	-	-	-	3	2	-	-
CO2	3	3	3	2	3	2	1	1	2	2	2	3	3	3	2
CO3	3	3	3	2	3	2	1	1	3	2	2	3	2	2	3
CO4	3	3	3	3	3	2	1	1	3	2	3	3	-	2	2

Syllabus

Unit 1

Overview of omics technologies (transcriptomics, proteomics, metabolomics) - history and evolution of omics technologies – the importance of application of AI omics research – the role of omics research in biological research.

Unit 2

Gene expression profiling using microarrays and RNA sequencing - transcriptomics data analysis using bioinformatics tool and AI-based methods - functional annotation of transcripts and pathways - alternative splicing analysis - single-cell transcriptomics and spatial transcriptomics.

Unit 3

Protein separation and identification using mass spectrometry - quantitative proteomics and label-free quantification - proteome analysis using bioinformatics tools and AI-based algorithm - proteins and pathways - protein-protein interactions and network analysis - structural proteomics and protein modifications.

Unit 4

Metabolite identification and quantification using mass spectrometry and NMR - metabolite profiling and metabolic pathway analysis - metabolomics data analysis using bioinformatics tools and AI-based algorithms - metabolic flux analysis and isotopic labelling - metabolomics in systems biology and synthetic biology - metabolomics in disease diagnosis and drug discovery.

Unit 5

Multi-omics data integration and analysis - correlation and causation analysis - AI-based predictive models - biomarker discovery and validation - A-based method to analyse biomarker data – the future direction of omics research with AI.

Text Books / References

"Transcriptomics and Gene Regulation" by Simon Tavaré and Nancy R. Zhang

"Proteomics: Methods and Protocols" edited by Daniel J. Liebler

"Metabolomics: From Fundamentals to Clinical Applications", edited by Ute Roessner and J. Cameron Thrash

"Integrated Omics Analysis: Methods and Applications" edited by Andrew J. Percy and David S. Wishart

"Transcriptome Analysis: Methods and Applications" edited by Vinita Chauhan, Rakesh Sharma, and Debmalya Barh

"Proteomics in Systems Biology: Methods and Protocols" edited by Daniel J. Kliebenstein

"Metabolomics in Practice: Successful Strategies to Generate and Analyze Metabolic Data" by Michael Lammerhofer and Wolfram Weckwerth Recent literature

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

xxxxxxx	High-Performance and Cloud Computing	L-T-P-C: 2- 0 -2- 3
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Course Objectives

- Familiarize student with architectural overview of modern HPC and GPU based heterogeneous architectures, focusing on its computing power versus data movement needs.
- Familiarise the students working with cloud platforms and services to configure and use computational resources and storage.
- To educate students how to write efficient parallel programming and GPU programming.
- To discuss various application of HPC computational techniques in computational science.

Course Outcomes

After completing this course, students will be able to

CO1	Apply high-performance computing in different research field.
CO2	Design OpenMPI programme and CUDA programme
CO3	Simulate on cloud computing system.
CO4	Evaluate how the convergence of HPC and AI is transforming the data science.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	2	-	2	2	3	2	-	-	2	1	3	1	1	-
CO2	2	3	3	3	3	-	1	-	2	1	2	3	3	3	1
CO3	2	2	3	2	3	1	1	-	2	1	2	3	2	2	1
CO4	2	2	1	2	3	2	1	1	2	1	2	3	3	3	3

Syllabus

Unit 1

Introduction to basic architecture and OS concepts - multi-core CPUs - high-speed interconnects - overview of High-Performance Computing (HPC) and its importance - hardware and software requirements for HPC - parallel programming and their applications (MPI/OpenMP) – brief introduction to workload manager and job schedulers.

Unit 2

Overview of GPU architecture and its evolution - comparison of CPU and GPU architecture - introduction to CUDA programming and its basic concepts - optimizing CUDA kernels for maximum performance - advanced CUDA programming techniques, such as shared memory, constant memory, and texture memory - introduction to cuBLAS and cuDNN libraries for linear algebra and deep learning - case studies of GPU: accelerated applications in scientific computing, data analytics, and machine learning.

Unit 3

Definition of cloud computing and its importance - Benefits and challenges of cloud computing - types of cloud services (IaaS, PaaS, SaaS) and their characteristics - cloud computing architecture and its components - cloud storage and its types - cloud networking and its challenges - cloud security and its importance - cloud application - benefits and challenges of HPC and AI - synergy between HPC and AI - training and inference of AI models using HPC.

Text Books / References

"High Performance Computing: Modern Systems and Practices" by Thomas Sterling and Matthew Anderson

"CUDA by Example: An Introduction to General-Purpose GPU Programming" by Jason Sanders and Edward Kandrot

"Parallel Programming with MPI" by Peter S. Pacheco

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2 Assignments)	Internal	20
Quiz (minimum 2 Quizzes)	Internal	20
Min Project	Internal	20
Mid. Term Periodical	Internal	10
End Semester Exam (Theory and Practical)	External	30

xxxxxxxx	Genomics Data Science	L-T-P-C: 2-0-2-3
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Course Objectives

- Provide a strong foundation in genomics.
- Teach computational and statistical tools for analysing large-scale genomics data, gene sequence and expression analysis.
- Familiarize students with the tools, algorithm, data structure and principles of contemporary genomics (DNA sequencing, cancer genomics, single-cell sequencing and next-generation sequencing etc.)
- Introduced Python and R for DNA sequencing.

Course Outcomes

After completing this course, students will be able to

CO1	Design sequence assembly in Genomics Data Science.
CO2	Develop the ability to apply advanced computational methods to analyse genomics data.
CO3	Develop the ability to identify the main results from a genomics research study and interpret figures from primary research papers.
CO4	Analyse genomics data and design simulation experiments and interpret results.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	3	3	3	2	1	-	2	-	1	3	3	-	-
CO2	3	3	3	3	3	2	1	-	2	-	1	3	3	3	2
CO3	3	3	3	3	3	2	-	-	2	-	-	3	2	3	2
CO4	3	3	3	3	3	2	1	-	2	-	-	3	-	2	3

Syllabus

Unit 1

Introduction to genomics – functional genomics – gene – different genomics experimental technology (next-generation sequencing, CRISPR/Cas9 etc.) – programming in R

Unit 2

Data cleaning and processing – sequence alignment for gene analysis – gene expression analysis – differential gene expression – Pathway analysis and functional annotation – machine learning in genomics.

Unit 3

Comparative genomics – epigenomics – metagenomics – pharmacogenomics – gene regulatory network analysis.

Unit 4

Precision medicine and personalised genomics – Cancer genomics and analysis – Infectious disease genomics and analysis

Text Books / References

“Bioinformatics: Sequence and Genome Analysis” by David Mount

“Genomics and Personalized Medicine: What Everyone Needs to Know” by Michael Snyder

“Practice Computing for Biologists” by Steven Haddock and Casey Dunn

“Bioinformatics and Functional Genomics” by Jonathan Pevsner

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

xxxxxxx	AI in system biology	L-T-P-C: 2- 0 -2- 3
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Course Objectives

- The goal of this is to cover the basic concept and principles of artificial intelligence and their application in field system biology.
- Explore AI and other computational methods to analyse biological networks and complex biological systems.
- Explore enzyme kinetics, metabolic pathway, flux balance analysis and signal transduction.
- Cover mathematical, statistical and dynamics modelling of biological networks.

Course Outcomes

After completing this course, students will be able to

CO1	Analysis biological networks
CO2	Develop the ability to apply ML/DL techniques to understand and analyse biological networks, signal transduction, metabolic network, and gene regulatory network in physiology and pathology.
CO3	Design and simulate network-based mathematical and statistical models.
CO4	Develop pipelines for biochemical pathway analysis based on recent Omics studies and AI developments.

CO-PO Mapping

[illegible]

PSO	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO															
CO1	3	3	2	2	3	-	-	-	3	2	3	3	3	-	-
CO2	3	3	3	3	3	2	-	-	3	2	3	3	-	3	2
CO3	3	2	3	3	3	-	-	-	3	2	3	3	-	3	2
CO4	3	3	3	2	3	-	-	-	3	2	3	3	-	2	3

Syllabus

Unit 1

Introduction to complex systems - cell signalling – metabolic network – experimental technologies - network building analysis - enzyme kinetics – metabolic analysis – flux balance analysis

Unit 2

Modelling, construction and simulation of the biological network using AI - dynamic modelling of biological systems – network structure analysis using AI– ML/DL algorithm to design gene circuit

Unit 3

Modelling the behaviour of single cells using AI – AI accelerated cell and gene therapy – AI to analyse cryo-electron tomogram.

Unit 4

Ethical and legal considerations of AI in System biology - Future directions and opportunities for AI in Biotechnology

Text Books / References

Raman K (2021) An Introduction to Computational Systems Biology: Systems-Level Modelling of Cellular Networks. 1/e ISBN 9781138597327 (Chapman and Hall/CRC)

Voit E (2012) A First Course in Systems Biology. Garland Science, 1/e. ISBN 081534467 Recent literature

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

xxxxxxx	AI in drug design	L-T-P-C: 2- 0 -2- 3
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Course Objectives

- To educate students on fundamentals stages of drug discovery pipeline and how computational and informatics techniques can accelerate the pace of drug discovery.
- To teach them how to encode a molecule into numerical molecular descriptors and strings, allowing computational treatment of molecules.

- Discuss drug-likeness, informatics approaches to the prediction of chemical properties - QSAR, pharmacophores, pharmacokinetics, and pharmacodynamics.
- To teach them how to apply AI / ML techniques and construct validated classification and regression models for biological endpoints.

Course Outcomes

After completing this course, students will be able to

CO1	Explain the drug discovery pipeline and the role of computational drug discovery in the process.
CO2	Construct SMILES Representations of Molecular Structures
CO3	Extract chemical information from Molecular Structures in different chemical file formats.
CO4	Compute different kinds of Molecular Descriptors and Fingerprints, and construct similarity kernels from them.
CO5	Construct validated Classification and Regression models for pharmacological endpoints and evaluate model performance and Domain of Applicability.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	-	-	2	2	3	-	1	1	1	-	-	2
CO2	2	3	3	-	2	-	-	-	-	1	-	1	2	-	-
CO3	2	3	3	-	2	-	-	-	-	1	-	1	2	-	-
CO4	3	2	2	2	3	-	-	1	1	-	-	2	2	2	-
CO5	2	1	3	3	3	-	-	1	1	1	-	2	3	3	2

Syllabus

Unit 1

Why Computational Drug Design? The Drug Discovery pipeline; Chemical Space; Cheminformatics and Virtual High Throughput Screening; Lipinski's Rules of 5 and ADMET Modelling; Pharmacodynamics and pharmacokinetics; Structure-Based and Ligand-Based Drug Design.

Unit 2

Chemical File Formats and Representations; Topological Indices; Substructural Descriptors and 2D fingerprints; Local Molecular Surface Property Descriptors; 3D Shape and Chiral descriptors; Molecular Similarity Measures and Kernels; Chemical and Biological Networks.

Unit 3

Linear Free Energy Relationships; Pharmacophores and Molecular Interaction Fields; Model Validation; Structure Based Methods - Docking & Scoring; Molecular Simulation (Monte-Carlo, Molecular dynamics) – Structure and dynamics properties (normal mode analysis) – Gaussian and Gauss view – Density functional theory.

Unit 4

Linear and Non-Linear Models; Classification, Regression and Ranking; Data preprocessing, Performance Measures and unbalanced datasets; Dimensionality reduction and Feature selection; Evolutionary computing; Kernel methods; Best Practices in Predictive Modeling; Applications of Deep Learning in Pharma.

Text Books / References

Johann Gasteiger, Thomas Engel, Chemoinformatics: A Textbook (Wiley-VCH, 2003)

Jürgen Bajorath (Editor), *Cheminformatics and Computational Chemical Biology (Methods in Molecular Biology)* (Humana Press, 2004)

Andrew R. & Leach, Valerie Gillet, *An Introduction to Chemoinformatics* (Springer International, New Delhi, 2009)

Merz Jr, K.M., Ringe, D. and Reynolds, C.H. (Editors) *Drug design: structure-and ligand-based approaches* (Cambridge University Press, 2010).

N. Sukumar, et al, Computational Drug Discovery: A Primer (Ion Cure Press, 2023).

John L. Lamattina, Drug Truths: Dispelling the Myths about Pharma R&D (John Wiley, Hoboken, NJ, 2008)

Barry Werth, *The Billion Dollar Molecule: One Company's Quest for the Perfect Drug* (Simon & Schuster, 1995)

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

xxxxxxx	Applied Cryptography	L-T-P-C: 2- 0 -2- 3
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Course Objectives

- To understand the underlying fundamentals of classical and modern cryptography
- To understand about the cryptographic systems used over the Internet.
- To help the students to study cryptographic protocols, common attacks and their prevention.

Course Outcomes

After completing this course, students will be able to

CO1	Analyze the concepts of classical and modern cryptography.
CO2	Analyze the common attacks and the preventive systems.
CO3	Apply appropriate cryptographic techniques to a security engineering problem
CO4	Implement standard security protocols.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															

CO1	3	3	2	2	-	-	-	-	2	2	-	3	2	2	1
CO2	3	3	3	3	-	-	-	-	2	2	-	3	1	2	1
CO3	3	2	3	3	1	-	-	-	2	2	-	3	2	2	1
CO4	3	2	3	3	-	-	-	-	2	2	-	3	1	2	1

Syllabus

Unit 1

Number theory concepts – Divisibility, GCD, modular exponential, congruence, Chinese remainder theorem. Groups, rings, fields.

Unit 2

Overview of Cryptography , Symmetric key cryptography, stream ciphers, block ciphers, DES and Enhancements, AES, Attacks on block ciphers, Message integrity- Message integrity: definition and applications. Hashing, collision resistance. Public key cryptography- Arithmetic modulo primes, Cryptography using arithmetic modulo primes, Public key encryption Arithmetic modulo composites, RSA, Attacks on RSA, Rabin Cryptosystem, Discrete Logarithm Problem and related Algorithms, ElGamal Cryptosystem

Unit 3

Introduction to Elliptic Curve Cryptography, Digital signatures: definitions and applications, More signature schemes and applications, Identification protocols, Authenticated key exchange and SSL/TLS session setup, Zero knowledge protocols. Key agreement protocols, Diffie-Hellman protocol, variations.

Text Books / References

James Strayer, Elementary Number Theory, Waveland Press, 2002.

Katz, Jonathan, and Yehuda Lindell. Introduction to modern cryptography. Chapman and Hall/CRC, 2014

Katz, Jonathan, Alfred J. Menezes, Paul C. Van Oorschot, and Scott A. Vanstone. Handbook of applied cryptography. CRC press, 1996.

Stallings, William. Cryptography and network security: principles and practice. Upper Saddle River:Pearson, 2017.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

xxxxxxx

Network and Wireless security

L-T-P-C: 2- 0 -2- 3

Course Objectives

- To understand the various techniques for network protection.

- To gain knowledge on security and privacy issues on various networks.
- To study the use of modern tools and techniques used in analysing various security parameters of the network.

Course Outcomes

After completing this course, students will be able to

CO1	Analyse the network security protocols and standards.
CO2	Evaluate the various network security threats.
CO3	Apply modern tools to simulate network attacks.
CO4	Interpret the ethical aspects of network security.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	2	2	3	-	-	-	2	2	-	3	1	2	1
CO2	3	3	3	3	3	-	-	-	2	2	-	3	2	2	1
CO3	3	2	3	3	3	-	-	-	2	2	-	3	2	2	1
CO4	3	2	3	3	3	-	-	-	2	2	-	3	1	2	1

Syllabus

Unit 1

Overview of network security principles, concepts, and terminology. Threats to network security, including attacks on confidentiality, integrity, and availability. Network security goals and objectives. IP tables, NAT. Intrusion detection systems, Types. Honeypots.

Unit 2

Network protocols and standards, SSL, TLS, IPsec, HTTPS. Wireless network architecture and protocols, Security risks and threats in wireless networks, Wireless security standards, including WPA and WPA2. Mobile security architecture, Mobile device security risks and threats, Mobile security standards, including iOS and Android.

Unit 3

Common network attacks. Wireshark, packet sniffing. LAN attacks, ARP cache spoofing, ettercap. Analysing network vulnerability using NMAP. Use of metasploit and metasploitable to perform common attacks over the network. Usage of aircrack-ng suite. Legal issues in network security, cyber crime and cyber law. Ethical issues in network security, privacy and intellectual property rights.

Text Books / References

William Stallings, Cryptography and Network Security: Principles and Practice, 8th Edition, Pearson edition, 2020.

Behrouz A. Forouzan, Cryptography & Network Security, McGraw-Hill, 3rd Edition 2015.

Jeremy Faircloth, Penetration Tester's Open Source Toolkit, 4th Edition, 2016.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

xxxxxxx	Intrusion Detection & Prevention Systems	L-T-P-C: 2- 0 -2- 3
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Course Objectives

- This course helps the students to understand the basics of intrusion detection and prevention systems, including concepts of intrusion detection, types of intrusions, vulnerabilities, and threats.
- This course helps the students to learn the different types of intrusions and their potential impacts.
- This course also provides methods to explore different techniques and methodologies for intrusion detection and prevention.
- This course will help students to gain practical experience with popular intrusion detection and prevention systems.

Course Outcomes

After completing this course, students will be able to

CO1	Analyze the characteristics of various network attacks and select appropriate intrusion detection and prevention techniques to mitigate the associated risks.
CO2	Configure open-source intrusion detection and prevention systems to detect and prevent network attacks
CO3	Implement machine learning and deep learning models for detecting intrusions
CO4	Analyze the effectiveness of intrusion detection and prevention systems

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	1	1	-	-	2	2	-	2	-	-	-	1	1	2	1
CO2	1	1	1	1	3	2	-	2	-	-	-	2	1	-	1
CO3	3	2	2	2	3	2	-	2	-	-	-	2	3	2	1
CO4	1	1	-	-	1	1	-	1	-	-	-	1	-	1	1

Syllabus

Unit 1

Introduction to Intrusion Detection and Prevention Systems - Overview of Intrusion Detection and Prevention Systems - Types of Intrusions - Goals and Requirements of Intrusion Detection and Prevention Systems - Differences between IDS and IPS

Unit 2

Network Security Fundamentals - Network Security Basics - OSI Reference Model - TCP/IP Protocol Suite - Network Security Threats - Network-Based IDS - NIDS Architecture - NIDS Detection Techniques - Snort: An Open Source NIDS - Snort Rules - Host-Based IDS - HIDS Architecture - HIDS Detection Techniques - OSSEC: An Open Source HIDS - OSSEC Rules - Anomaly Detection - Anomaly Detection and Classification - Machine Learning Techniques for Anomaly Detection

Unit 3

Intrusion Prevention Systems - IPS Architecture and Mechanisms - Signature-Based IPS - Behavior-Based IPS

Text Books / References

Ali A. Ghorbani, Wei Lu, "Network Intrusion Detection and Prevention: Concepts and Techniques", Springer, 2010.

Carl Enrolf, Eugene Schultz, Jim Mellander, "Intrusion detection and Prevention", McGraw Hill, 2004

Paul E. Proctor, "The Practical Intrusion Detection Handbook ", Prentice Hall, 2001.

Ankit Fadia and Mnu Zacharia, "Intrusiion Alert", Vikas Publishing house Pvt., Ltd, 2007.

Earl Carter, Jonathan Hogue, "Intrusion Prevention Fundamentals", Pearson Education, 2006.

Rafeeq Ur Rehman. Intrusion detection systems with Snort: advanced IDS techniques using Snort, Apache, MySQL, PHP, and ACID. Prentice Hall Professional, 2003.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

xxxxxxx	Software Vulnerability Analysis	L-T-P-C: 2- 0 -2- 3
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Course Objectives

- This course teaches software engineering techniques for building security into software as it is developed.
- This course introduces students to the discipline of designing, developing, and testing secure and dependable software-based systems.
- This course provides hands on experience in software security analysis and development
- This course helps students to learn how to mitigate software vulnerabilities through secure software development practices

Course Outcomes

After completing this course, students will be able to

CO1	Analyse the security risk of a system under development.
CO2	Apply secure coding practices to prevent common vulnerabilities from being injected into software.
CO3	Design security requirements (which include privacy requirements).

CO4	Validate security requirements.
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CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	2	1	2	1	2	2	-	2	-	-	-	2	-	2	1
CO2	1	2	2	3	3	2	-	1	-	-	-	2	1	1	-
CO3	1	2	3	2	3	2	-	1	-	-	-	2	-	1	-
CO4	-	1	-	-	1	2	-	2	-	-	-	2	-	2	-

Syllabus

Unit 1

Introduction to Software Vulnerability Analysis - Overview of software vulnerabilities - Importance of vulnerability analysis in software development - Types of Software Vulnerabilities

Unit 2

Common types of software vulnerabilities - Code Injection vulnerabilities - Authentication and Authorization vulnerabilities - Input Validation vulnerabilities - Static Analysis - Source code analysis techniques - Dynamic Analysis - Types of dynamic analysis techniques - Binary Analysis - Overview of binary analysis - Reverse engineering techniques

Unit 3

Mitigation Strategies - Overview of software security mitigation strategies - Secure coding practices - Input validation and output encoding - Security testing and verification techniques - Security standards and best practices

Text Books / References

Anley, Chris, John Heasman, Felix Lindner, and Gerardo Richarte. *The shellcoder's handbook: discovering and exploiting security holes*. John Wiley & Sons, 2011.

Takanen, Ari, Jared D. Demott, Charles Miller, and Atte Kettunen. *Fuzzing for software security testing and quality assurance*. Artech House, 2018.

Sikorski, Michael, and Andrew Honig. *Practical malware analysis: the hands-on guide to dissecting malicious software*. no starch press, 2012.

Dowd, Mark, John McDonald, and Justin Schuh. *The art of software security assessment: Identifying and preventing software vulnerabilities*. Pearson Education, 2006.

McGraw, Gary. "Software security." *Building security in*, Addison-Wesley Professional, 2006.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30

Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

19AIEXXX	Cybercrime Investigations & Digital Forensics	L-T-P-C: 1-2-0-3
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Course Objectives

- To provide overview of global reach of the Internet and various cybercrimes in various domains.
- This course provides an overview of cybercrime and the digital law enforcement practices put in place to respond to them.
- The course will focus on the types and extent of current cyber-crimes, how the justice system responds to these crimes, the various constitutional protections afforded to computer users, the law and policies that govern cybercrime detection and prosecution, and related technologies.

Course Outcomes

After completing this course, students will be able to

CO1	Enable the student to define and describe the nature and scope of cybercrime.
CO2	Develop knowledge of major incidents of cybercrime and their resulting impact.
CO3	Facilitate the student to analyse and discuss national and global digital law enforcement efforts
CO4	Evaluate the specific AI and ML enabled technology that facilitates cybercrime and digital law enforcement

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	2	1	2	2	3	3		3		1		3			2
CO2	2	2	3	2	3	3		3		2		3			2
CO3	2	3	3	2	3	3	2	3	3	2	2	3		2	2
CO4	3	1	3	2	3	2	1	3	2	1	1	1		2	3

Syllabus

Introduction to cybercrime, criminal law, courts, and lawmaking, Types of computer-related crimes, Sources of cybercrime law (substantive and procedural), Technology, cybercrime, and police investigations, Technology and crime, Cyber deviance, cybercrime, and cyber terror, Computer misuse crimes, Malware and automated computer attacks, Malware, DDoS attacks, and Botnets, Digital piracy and Intellectual property theft, Digital piracy, Copyright, trademark, and trade secrets, Pornography, prostitution, and sex crime, The Fourth Amendment, computers, and computer networks, Digital/Computer Forensics -Introduction to digital and computer forensics, Legal issues related to digital investigations, National security and international.

Text Books / References

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															

CO1	2	1	3	2	3	2	2		3	2	1	3			
CO2	2	2	3	2	3	2	2	1	3	2	1	3			
CO3	2	3	3	2	3	2	2	1	3	2	1	3			
CO4	3	2	2	2	2	2	2	3	2	1	1	2			

Syllabus

Understanding the Core Concepts of Distributed Systems -distributed systems designs, system constraints, trade-offs and techniques in distributed systems, distributed system for different data and applications, Distributed system security-Access and location transparency, Processes and Communication, naming, Parallelization of tasks - Concurrency and Synchronization, Consistency and Replication, Distributed system Security and network protocols – types of attacks, encryption algorithms, authentication, public key cryptosystems, data verification.

Text Books / References

Anderson, Ross. Security engineering: a guide to building dependable distributed systems. John Wiley & Sons, 2020.

Andrew S. Tannenbaum and Maarten Van Steen, “Distributed Systems: Principles and Paradigms”, Second Edition, Pearson, 2007.

Belapurkar, Abhijit, Anirban Chakrabarti, HarigopalPonnapalli, Niranjana Varadarajan, Srinivas Padmanabhuni, and Srikanth Sundarajan. Distributed systems security: issues, processes and solutions. John Wiley & Sons, 2009.

George Coulouris, Jean Dollimore, Tim Kindberg, and Gordon Blair, “Distributed Systems: Concepts and Design”, Fifth Edition, Addison Wesley, 2011.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Big Data Analytics in Finance	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- This course introduces students to the use of big data analytics in finance.
- Students will learn to use various approaches and tools required to collect, process, analyse, and understand financial data.
- The course will cover data mining, machine learning, statistical analysis, and data visualisation, with an emphasis on financial applications.
- Students will receive hands-on expertise with common data analytics software tools and programming languages.

Course Outcomes

After completing this course, students will be able to

CO1	Understand the fundamentals of big data analytics and its applications in finance
CO2	Collect, clean, and process financial data for analysis
CO3	Apply various analytical techniques to financial data, including data mining, machine learning, and statistical analysis
CO4	Critically evaluate the ethical and legal implications of big data analytics in finance

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	2	2	2	3	-	-	-	3	2	3	3	-	-	-
CO2	3	3	3	3	3	-	-	-	3	2	3	3	-	-	-
CO3	3	3	3	3	3	-	-	-	3	2	3	3	-	-	-
CO4	3	3	3	3	3	-	-	-	3	2	3	3	-	-	-

Syllabus

Unit 1

Introduction to Big Data Analytics in Finance, Data Collection and Pre-processing, Exploratory Data Analysis and Visualization

Unit 2

Statistical Analysis for Financial Data, Machine Learning Techniques for Financial Data, Financial Time Series Analysis, Fraud Detection and Prevention

Unit 3

Risk Management using Big Data Analytics, Trading and Investment using Big Data Analytics, Regulatory Compliance and Legal Issues in Big Data Analytics

Textbook / Reference

Pries, K.H., & Dunnigan, R. (2015). Big Data Analytics: A Practical Guide for Managers.

Big Data Science in Finance (Wiley, 2021), Irene Aldridge and Marco Avellaneda

Big Data and Machine Learning in Quantitative Investment, Tony Guida (Wiley, 2018)

Handbook of Financial Risk Management: Simulations and Case Studies, Ngai Hang Chan, Hoi Ying Wong (Wiley, 2013)

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20

Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Blockchain Technologies	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- This course addresses the fundamentals of Blockchain technology and how blockchain systems (mainly Bitcoin and Ethereum) work.
- It provides students a knowledge of essential ideas and developments in the world of cryptocurrencies and distributed ledger systems.
- Integrate ideas from blockchain technology into their own projects

Course Outcomes

After completing this course, students will be able to

CO1	Design, build, and deploy Bitcoin
CO2	Implement smart contract in real world applications.
CO3	Make decisions about the use (or not) of blockchain technology in systems, and support decisions with relevant arguments.
CO4	Apply various blockchain concepts to analyze examples, proposals, case studies, and preliminary blockchain system design discussions.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	3	3	3	3	3	-	-	-	3	3	-	3			
CO2	3	3	3	2	3	-	-	-	3	3	-	3			
CO3	3	3	3	3	3	-	-	-	3	3	-	3			
CO4	3	2	3	3	3	-	-	-	3	3	-	3			

Syllabus

Unit I

Blockchain Overview and Introduction, History and Origin of Blockchain (and Cryptocurrency), Technical Concepts of Blockchain Systems, Digital Security Technology, Decentralized networks and technology (serverless)

Unit 2

Key Problem Challenges and Solutions, Ownership, Transaction Records (Ledger) Incentives, Attacks, (Distributed) Consensus, how traditional Electronic Payment Systems work, DLT Technical Concepts Continued, Electricity Use, Bitcoin blockchain, Challenges and solutions, proof of work, Proof of stake, alternatives to Bitcoin consensus, Bitcoin scripting language and their use

Unit 3

Text Books / References

Mastering Ethereum: Building Smart Contracts and DApps, Andreas M. Antonopoulos, Dr. Gavin Wood, Gavin Wood, O'Reilly Media, Incorporated, 2018

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

***** Speech Processing L-T-P-C: 2- 0- 3- 3

Course Objective

- To understand the principles of speech processing, human speech production and perception system.
- To estimate excitation and vocal tract features using time and frequency domain processing techniques.
- To explore the various conventional, machine learning and deep learning models for speech classification, recognition, synthesis, and detection tasks

Course Outcomes

After completing this course, students will be able to

CO1	Analyse the acoustic/articulatory characteristics of different speech regions and speech sounds
CO2	Apply time and frequency domain processing techniques to speech signals
CO3	Analyse and extract relevant spectral parameters and temporal parameters of speech signal
CO4	Evaluate the performance of a model or algorithm (conventional/Machine learning/Deep learning) developed for a speech technology application

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
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CO															
CO1	3	3	3	3	2	1	1	1	2	3	1	2	1	1	1
CO2	3	3	3	3	3	1	1	1	2	2	1	2	3	2	1
CO3	3	3	3	3	3	1	1	1	2	2	1	2	3	2	2
CO4	3	3	3	3	3	2	1	2	2	2	1	3	3	3	3

Syllabus

Unit 1

Introduction to Speech Processing, Overview of the human speech production system, acoustic and physiological mechanisms of speech production, glottal signal characteristics and source features, significance of glottal activity regions, speech signal characteristics, acoustic/articulatory characteristics of different speech sounds -vowels and consonants.

Unit 2

Short time processing of speech for estimation of excitation and vocal tract features - Time Domain processing- Energy, magnitude, zero crossing rate, STACF, Linear Prediction Analysis, Frequency domain processing and Spectro-temporal representation of speech signal- Narrowband, wideband spectrograms, Cepstral Analysis, Melspectrogram, MFCC feature extraction.

Unit 3

Speech data preparation and feature engineering, machine learning versus deep learning models in speech classification tasks (age, gender, dialect/accent), Automatic speech recognition (ASR) - statistical models- Hidden Markov Models (HMMs) for ASR, Deep learning speech recognition pipeline (end-to-end models), overview of other speech technology applications such as emotion recognition, speaker recognition, speech synthesis, and speech pathology detection.

Text Books / References

'Fundamentals of Speech Recognition', L. Rabiner, Biing-Hwang Juang and B. Yegnanarayana, Pearson Education Inc. 2009

'Speech Communication', Douglas O'Shaughnessy, University Press, 2001

'Discrete Time Speech Signal Processing', Thomas F Quatieri, Pearson Education Inc., 2004

Hannun, Awni, et al. "Deep speech: Scaling up end-to-end speech recognition." *arXiv preprint arXiv:1412.5567* (2014).

Collobert, Ronan, Christian Puhresch, and Gabriel Synnaeve. "Wav2letter: an end-to-end convnet-based speech recognition system." *arXiv preprint arXiv:1609.03193* (2016).

Gulati, Anmol, et al. "Conformer: Convolution-augmented transformer for speech recognition." *arXiv preprint arXiv:2005.08100* (2020).

Shen, Jonathan, et al. "Natural TTS synthesis by conditioning wavenet on mel spectrogram predictions." *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2018*

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Text Analytics	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- The main objective of the course is to understand the leading trends and systems in Natural Language Processing.
- This course will help the students to understand the basic representations used in syntax, the semantics of Natural Language Processing.
- This course will help the students to understand and explore the models used for word/sentence representations for various NLP applications.
- This course will help the students to implement deep learning algorithms in Python and learn how to train deep networks for NLP applications.

Course Outcomes

After completing this course, students will be able to

CO1	Apply modern tools for solving problems in computational linguistics
CO2	Implement word representation models to solve NLP problems
CO3	Develop machine learning/deep learning models for solving NLP applications
CO4	Evaluate the performance of NLP models

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	1	2	1	2	3	-	-	1	-	-	-	2	1	-	1
CO2	2	2	2	2	3	1	-	1	-	-	-	2	3	2	1
CO3	3	2	2	2	3	1	-	1	-	-	-	2	3	2	1
CO4	-	-	-	1	2	1	-	1	-	-	-	1	-	2	1

Syllabus

Unit 1

Computational linguistics- Introduction, syntax, semantics, morphology, collocation and other NLP problems.

Unit 2

Word representation: One-hot encoding, Bag-of-Words (BoW) Dictionary: Term Frequency – Inverse Document Frequency (TF-IDF), Language Model-n-gram – Neural Network-based word embedding algorithms

Unit 3

Sequences and sequential data: Machine learning and deep learning for NLP, Sequence to sequence modelling - BERT, GPT, Graph NLP, Hidden Markov Model, Conditional Random Field, Topic modelling

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															

CO1	3	3	3	3	3	3	1	-	3	3	2	3	3	1	1
CO2	3	3	3	2	3	2	-	-	3	3	2	3	3	2	2
CO3	3	3	3	3	2	1	-	-	3	3	3	3	3	3	3
CO4	3	3	3	3	2	1	-	-	3	3	3	3	2	3	3

Syllabus

Unit 1

Goals and history of IR, impact of web in IR, basic concepts- query, document, corpus, text representation and evaluation, Boolean model, TF-IDF, vector-space retrieval models, Probabilistic retrieval models

Unit 2

Text similarity metrics, Tokenizing, language models, KL-divergence, performance metrics, reference collections and evaluation of IR systems, query languages for IR, relevance feedback, query expansion-local and global

Unit 3

Web search, web crawling, link analysis – hits, page rank, matrix decompositions and latent semantic indexing, Deep learning for IR- word embeddings, neural language models

Text Books / References

Christopher D. Manning, Prabhakar Raghavan and Hinrich Schütze, Introduction to Information Retrieval, Cambridge University Press. 2008. <http://nlp.stanford.edu/IR-book/information-retrieval-book.html>

ChengXiang Zhai, Statistical Language Models for Information Retrieval (Synthesis Lectures Series on Human Language Technologies), Morgan & Claypool Publishers, 2008.

Stefan Buettcher, Charles L. A. Clarke, Gordon V. Cormack, Information Retrieval: Implementing and Evaluating Search Engines, MIT Press. (2010)

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx

Introduction to Chatbots

L-T-P-C: 2- 0- 2- 3

Course Objectives

- This course helps the students to develop a functional chatbot that can respond to user queries and complete simple tasks.
- Apply NLP techniques to improve the chatbot's ability to understand and respond to user input.

- Apply appropriate evaluation metrics to assess the performance of a chatbot.
- Evaluate the performance of a chatbot and make necessary adjustments to improve its functionality.
- Critically analyze and evaluate the ethical implications of using chatbots in various contexts, and propose ethical solutions to address these issues.

Course Outcomes

After completing this course, students will be able to

CO1	Implement chatbots with different frameworks
CO2	Develop machine learning/deep learning models for chatbots
CO3	Apply Natural Language Processing techniques to enhance chatbot capabilities
CO4	Evaluate the performance of chatbots using various metrics and techniques.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	2	1	2	1	3	2	-	1	-	-	-	2	1	-	1
CO2	2	2	2	3	3	2	-	2	-	-	-	2	3	1	1
CO3	1	1	1	2	3	2	-	2	-	-	-	2	3	1	1
CO4	-	-	-	1	2	1	-	1	-	-	-	1	-	2	1

Syllabus

Unit 1

Introduction to chatbots - definition and characteristics of chatbots - brief history of chatbots - types of chatbots
- Use cases and applications

Unit 2

Natural Language Processing (NLP) for Chatbots - Understanding language processing - Syntax and Semantics
- Pre-processing and Tokenization - Fundamental Methods of NLP for Building Chatbots

Unit 3

Machine Learning/Deep Learning for Chatbots - Machine Learning/Deep Learning Algorithms used in chatbots
- Data collection and annotation - Training and evaluation of chatbots - Designing and Implementing Chatbots -
Designing conversational interfaces - Implementing chatbots using frameworks like Dialogflow, Rasa, etc. -
Integration with different messaging platforms

Unit 4

Ethics in Chatbot Development - Responsible AI and ethics - Bias and fairness considerations in chatbot
development - Human-in-the-loop approaches for chatbot development

Text Books / References

Sumit Raj, Building Chatbots with Python Using Natural Language Processing and Machine Learning, First Edition, Apress Berkeley, CA, 2019 (Softcover ISBN 978-1-4842-4095-3).

Srini Janarthanam. *Hands-on chatbots and conversational UI development: build chatbots and voice user interfaces with Chatfuel, Dialogflow, Microsoft Bot Framework, Twilio, and Alexa Skills*. Packt Publishing Ltd, 2017.

Shevat, Amir. *Designing bots: Creating conversational experiences*. " O'Reilly Media, Inc.", 2017.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Machine Translation and Sequence-to-Sequence Models	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- The main objective of the course is to obtain basic understanding and implementation skills for modern methods for machine translation
- This course introduces different approaches to build machine translation systems
- This course helps the students to understand various evaluation metrics used for assessing the performance of a machine translation model
- This courses introduces different deep learning architectures used for implementing the machine translation system.

Course Outcomes

After completing this course, students will be able to

CO1	Implement a statistical machine translation system
CO2	Implement a neural machine translation system using RNN-based encoder-decoder architecture
CO3	Implement a neural machine translation system using transformer-based encoder-decoder architecture
CO4	Evaluate the performance of machine translation models

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	2	2	3	2	3	-	-	-	1	-	1	1	3	1	1
CO2	2	2	3	2	3	-	-	-	1	-	1	1	3	1	1
CO3	2	2	3	2	3	-	-	-	1	-	1	1	3	1	1
CO4	-	-	1	-	1	-	-	-	-	-	-	1	-	3	-

Syllabus

Introduction - Machine Translation Overview – Language Models – Rule based Machine Translation – Statistical Machine Translation – Encoder-decoder models – Attention mechanism - Neural Machine

Translation – Phrase based models – Tree based models – Subword level models – Transformer networks - Evaluation metrics

Text Books / References

*Daniel Jurafsky, James H Martin, Speech & language processing, preparation [cited 2020 June 1]
Available from: <https://web.stanford.edu/~jurafsky/slp3> (2018).*

Philipp Koehn, Statistical machine translation. Cambridge University Press, 2009..

Philipp Koehn, Neural machine translation. Cambridge University Press, 2020.

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Speech Recognition and Understanding	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- The main objective of the course is to give an introduction to recognition and understanding of the speech
- This course introduces different approaches to build a speech recognition system
- This course helps the students to understand language model and acoustic models required for building a speech recognition system
- This courses introduces different deep learning architectures used for implementing the end to end automatic speech recognition system

Course Outcomes

After completing this course, students will be able to

CO1	Implement language and acoustic models required for a speech recognition system
CO2	Develop a feature extraction model for speech recognition
CO3	Implement a speech recognition model
CO4	Develop a deep learning based end to end speech recognition model

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	2	2	2	2	2	-	-	1	-	-	-	2	3	1	1
CO2	2	2	2	2	2	-	-	1	-	-	-	2	3	1	1
CO3	2	2	2	2	2	-	-	1	-	-	-	2	3	1	1
CO4	2	2	2	2	2	-	-	1	-	-	-	2	3	1	1

Syllabus

Introduction of speech recognition - Overview of speech recognition systems - Speech recognition formulations - Feature extraction – Alignment – Hidden Markov Model - Advanced acoustic modelling - Language model - Deep learning for speech recognition - Advanced neural network architectures for acoustic model - End-to-end ASR

Text Books / References

L. R. Rabiner, B. H. Juang and B. Yegnanarayana, "Fundamentals of speech recognition", Pearson Education, 2009.

J. R. Deller, Jr., J. H. L. Hansen and J. G. Proakis Discrete-Time Processing of Speech Signals, Wiley-IEEE Press, NY, USA, 1999.

D. O'Shaughnessy, Speech Communications: Human and Machine, Second Edition, University Press, 2005.

J. Benesty, M. M. Sondhi and Y. Huang, "Handbook of speech processing", Springer, 2008.

*Daniel Jurafsky, James H Martin, Speech & language processing, preparation [cited 2020 June 1]
Available from: <https://web.stanford.edu/~jurafsky/slp3> (2018).*

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Social Media Text Analysis	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- The main objective of the course is to give an introduction to the key concepts about social media text data
- This course introduces machine/deep learning and topic modelling approaches to build a social media text classification model
- This course helps the students to monitor consumers and competitors and glean deeper consumer insights based on advanced social media data modelling
- This courses helps the students to make better business decisions by leveraging social media dat

Course Outcomes

After completing this course, students will be able to

CO1	Analyze social media text data
CO2	Implement a sentiment analysis model using machine/deep learning algorithms
CO3	Implement a topic model for social media text categorization
CO4	Analyze a social network

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
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CO																
CO1	-	-	-	1	2	1	-	1	-	-	-	1	-	2	-	
CO2	2	2	2	2	2	-	-	1	-	-	-	2	3	2	1	
CO3	2	2	2	2	2	-	-	1	-	-	-	2	3	2	1	
CO4	1	1	1	2	2	1	-	1	-	-	-	2	1	-	1	

Syllabus

Introduction to Social Media Analytics - Social media data collection - Processing unstructured and semi-structured data – Monitoring user engagement in social media - Identifying Opinions through Sentiment Analysis and Topic Modeling - Social Network Analysis and Metrics - Identifying Influencers in Social Network

Text Books / References

Szabo, G., G. Polatkan, O. Boykin & A. Chalkiopoulos, Social Media Data Mining and Analytics, Wiley, ISBN 978-1-118-82485-6, 2019

Golbeck, Jennifer. Analyzing the social web. Newnes, 2013.

Finger, L. & Dutta, S., Ask, Measure, Learn: Using Social Media Analytics to Understand and Influence Customer Behavior. Sebastopol, CA: O'Reilly, 2014

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Power System Stability	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- The course shall teach the students to analyse power system stability using both traditional and data-driven techniques.
- The course aims to enable students to understand and develop control strategies that can improve power system stability.
- By the end of the course, students should be able to simulate and analyse power system stability using synchronous machine models.
- The course intends to equip students with the critical thinking and problem-solving skills needed to tackle power system stability challenges.

Course Outcomes

After completing this course, students will be able to

CO1	Comprehend the fundamentals of power system stability
CO2	Apply mathematical modelling techniques to simulate power system stability

CO3	Familiarize and compare different techniques for improving power system stability
CO4	Apply data-driven techniques for power system stability analysis and improvement

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	2	2	-	1	1	1	2	-	-	-	-	2	1	-	-
CO2	2	3	2	2	2	1	2	-	-	-	1	2	3	-	2
CO3	2	3	2	2	2	1	1	-	-	-	1	2	2	-	2
CO4	2	3	3	2	2	1	2	-	-	-	1	2	3	-	3

Syllabus

Unit 1

Introduction to Power System Stability: Definition, types, and causes of power system instability - Overview of power system modelling - Synchronous generator models - Excitation system models - Governor models - Load models.

Unit 2

Power System Stability Analysis: Small-signal stability analysis - Transient stability analysis - Critical clearing time - Voltage stability - Basics of power system control

Unit 3

Improving Power System Stability: Power System Stabilizers - FACTS and HVDC systems for power system stability - Microgrids for power system stability - Data-driven Techniques for power system stability

Text Books / References

Prabha Kundur, Power System Stability and Control, 1998

John Grainger and William D. Stevenson Jr., Power System Analysis 1994

Harry G. Kwatny, Karen Miu-Miller, Power System Dynamics and Control, 2016

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Microgrids	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- To enable students to identify and analyse the benefits and hurdles of Microgrids in various industries and applications.
- To equip students with the ability perform comparative analysis of different control strategies and energy management systems.
- To develop the skillset towards applying optimization techniques for modelling and analysing Microgrids to achieve optimal performance.
- To develop the knowledge and proficiency for incorporating AI and ML-based to Microgrid optimization and control problems

Course Outcomes

After completing this course, students will be able to

CO1	Gain a critical comprehension of Microgrid technology, comprising its structure, constituents, and usage.
CO2	Acquaint themselves with various control tactics and energy management systems for Microgrids.
CO3	Utilize optimization methods to design and evaluate Microgrids for attaining optimal performance
CO4	Assess and incorporate advanced areas in Microgrid optimization and control, involving artificial intelligence and machine learning, to enhance Microgrid effectiveness and dependability.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	2	2	-	1	1	1	2		-	-		2	1	-	1
CO2	2	3	2	2	2	1	2		-	-	1	2	2	-	1
CO3	2	3	2	2	2	1	1		-	-	1	2	3	-	3
CO4	2	3	3	2	2	1	2		-	-	1	2	3	1	3

Syllabus

Unit 1

Introduction to Microgrids: Introduction - Overview of Microgrid architecture and components - Advantages and challenges of Microgrids - Applications of Microgrids in different industries

Unit 2

Microgrid Control : Control strategies for Microgrids - Energy management systems for Microgrids - Microgrid stability analysis and control - Microgrid protection and islanding

Unit 3

Microgrid Optimization : Microgrid modeling for optimization - Linear and nonlinear programming for Microgrid optimization - Convex optimization for Microgrid optimization

Unit 4

Advanced Topics in Microgrid Optimization and Control :Distributed optimization for Microgrids - Stochastic optimization for Microgrids - Artificial intelligence and machine learning for Microgrid control and optimization - Integration of renewable energy sources in Microgrids

Text Books / References

Nikos Hatziargyriou, Hassan Bevrani, and Jacob Ostergaard, "Microgrids: Control and Operation",2017

S. Chowdhury, S. P. Chowdhury, P. Crossley, "Microgrids and Active Distribution Networks",2009

Jizhong Zhu, "Optimization of Power System Operation",2015

Weerakorn Ongsakul and Vo Ngoc Dieu, "Artificial Intelligence in Power System Optimization",2013

Haitham Abu-Rub, Mariusz Malinowski, and Kamal Al-Haddad, "Power Electronics for Renewable Energy Systems, Transportation, and Industrial Applications",2014

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

23xxxxxx	Intelligent Energy Management Systems	L-T-P-C: 2- 0- 2- 3
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Course Objectives

- The course aims to help the students to identify and describe the fundamental concepts and principles of intelligent energy management.
- Students will learn to differentiate and categorize the key components of intelligent energy management systems, including sensors, controllers, and algorithms.
- The course will enable students to apply data analysis techniques to extract insights and trends from energy usage data and propose opportunities for energy savings and efficiency improvements.
- Students will learn to evaluate and compare potential benefits and challenges associated with implementing intelligent energy management in different contexts.

Course Outcomes

After completing this course, students will be able to

CO1	Identify and explain the key components of intelligent energy management systems and their application in optimizing energy consumption, using data analytics tools and techniques
CO2	Analyse energy usage data using various data analytics techniques to identify opportunities for energy savings and efficiency improvements and develop recommendations for implementation.
CO3	Evaluate the benefits and challenges of implementing intelligent energy management systems in buildings and other applications, using quantifiable metrics and data-driven approaches
CO4	Formulate simple intelligent energy management systems, using data analytics tools and techniques to optimize energy consumption and efficiency.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	2	1	-	1	-	1	2	-	-	-	-	1	1	-	1
CO2	2	3	2	2	2	1	2	-	-	-	1	2	2	-	2
CO3	2	2	2	2	2	1	3	-	-	-	1	2	3	-	3
CO4	3	3	3	3	2	1	2	-	-	-	1	2	3	-	3

Syllabus

Unit 1

Introduction to Energy Management: Overview - Types of energy management systems - Key components of intelligent energy management systems – Types of controllers and algorithms used in energy management systems - Algorithms for energy management and optimization - Data analytics techniques for energy management.

Unit 2

Energy Efficiency: Concepts of energy efficiency - Techniques for improving energy efficiency - Measurement and verification of energy savings using data analytics.

Unit 3

Demand Response: Introduction - Techniques for implementing demand response programs - Data-driven approaches to demand response.

Unit 4

Renewable Energy Integration & Storage: Challenges and opportunities for renewable energy integration with data-driven approaches - Overview of energy storage technologies - Integration of energy storage into energy management systems - Benefits and challenges of energy storage with data-driven approaches.

Text Books / References

Craig B. Smith and Kelly Parmenter, "Energy Management Principles, Second Edition", 2015

James Momoh, "Smart Grid: Fundamentals of Design and Analysis". 2012

Lawrence E. Jones, "Renewable Energy Integration: Practical Management of Variability, Uncertainty, and Flexibility in Power Grids", 2017

Pengwei Du and Ning Lu, "Energy Storage for Smart Grids: Planning and Operation for Renewable and Variable Energy Resources (VERs)", 2014

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

Course Objectives

- The course aims to enable students to understand the fundamental concepts of power quality management.
- The course shall empower the students to identify the various sources of power quality problems.
- The course aims to equip students with the ability to analyse power quality data and draw conclusions about the nature of the issues.
- Finally, the course shall enable students to develop data-driven solutions for power quality problems.

Course Outcomes

After completing this course, students will be able to

CO1	Identify and diagnose common power quality problems, their sources, effects, and significance.
CO2	Analyse the impact of various power quality problems on electrical systems and equipment.
CO3	Evaluate different power quality improvement techniques, towards the design optimal solutions
CO4	Interpret power quality data and formulate data-driven solutions for the quality improvement.

CO-PO Mapping

PO/ PSO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO															
CO1	2	3	1	3	2	1	2	-	-	-	-	2	2	-	2
CO2	2	2	1	3	2	1	2	-	-	-	1	2	1	-	1
CO3	2	3	2	3	2	1	1	-	-	-	2	2	3	-	2
CO4	3	3	3	3	2	1	2	-	-	-	1	2	3	-	3

Syllabus

Unit 1

Introduction to Power Quality Management: Definition and importance of power quality - Power quality terminology and units of measurement - Common power quality problems and their sources - Effects of power quality problems on electrical systems and equipment - Power quality indices and their significance – Power quality standards - Overview of power quality measurements and monitoring

Unit 2

Power Quality Problems: Harmonics and their causes - Harmonic distortion and its impact on electrical systems and equipment - Filtering techniques for mitigating harmonics – Voltage sags and swells: Overview, causes, effects and mitigation - Transient over- and under-voltages: Overview, causes, effects and mitigation

Unit 3

Power Quality Improvement Techniques: Overview and comparison of different techniques and their effectiveness - Reactive Power Compensation: overview and design - Active Power Filtering: overview and design – Energy storage for power quality improvement - Analysis and interpretation of power quality data.

Text Books / References

Ewald F. Fuchs and Mohammad A.S Masoum, “Power Quality in Power Systems and Electrical Machines” .Pearson Education, 2008

Bhim Singh, Ambrish Chandra, and Kamal Al-Haddad , “Power Quality Problems and Mitigation Techniques” , 2014

Jos Arrillaga, Bruce C. Smith, Neville R. Watson, and Alan R. Wood, “Power System Harmonic Analysis”, 2013

J. Benesty, M. M. Sondhi and Y. Huang, “Handbook of speech processing”, Springer, 2008.

*Daniel Jurafsky, James H Martin, Speech & language processing, preparation [cited 2020 June 1]
Available from: <https://web.stanford.edu/~jurafsky/slp3> (2018).*

Evaluation Pattern

Assessment	Internal/External	Weightage (%)
Assignments (minimum 2)	Internal	30
Quizzes (minimum 2)	Internal	20
Mid-Term Examination	Internal	20
Term Project/ End Semester Examination	External	30

FREE ELECTIVE

FREE ELECTIVES OFFERED UNDER MANAGEMENT STREAM

Cat.	Code	Title	L T P	Credit
HUM	19MNG331	Financial Management	3 0 0	3
HUM	19MNG332	Supply Chain Management	3 0 0	3
HUM	19MNG333	Marketing Management	3 0 0	3
HUM	19MNG334	Project Management	3 0 0	3
HUM	19MNG335	Enterprise Management	3 0 0	3
HUM	19MNG338	Operations Research	3 0 0	3
HUM	19MEE401	Industrial Engineering	3 0 0	3
HUM	19MEE346	Managerial Statistics	3 0 0	3
HUM	19MEE347	Total Quality Management	3 0 0	3
HUM	19MEE342	Lean Manufacturing	3 0 0	3
HUM	19CSE358	Software Project Management	3 0 0	3
HUM	19CSE359	Financial Engineering	3 0 0	3
HUM	19CSE360	Engineering Economic Analysis	3 0 0	3
HUM	19MNG331	Financial Management	3 0 0	3
HUM	19CSE362	Information Systems	3 0 0	3

FREE ELECTIVES OFFERED UNDER HUMANITIES / SOCIAL SCIENCE STREAMS				
Cat.	Code	Title	L T P	Credit
HUM	19CUL230	Achieving Excellence in Life - An Indian Perspective	2 0 0	2
HUM	19CUL231	Excellence in Daily Life	2 0 0	2
HUM	19CUL232	Exploring Science and Technology in Ancient India	2 0 0	2
HUM	19CUL233	Yoga Psychology	2 0 0	2
HUM	19ENG230	Business Communication	1 0 3	2
HUM	19ENG231	Indian Thought through English	2 0 0	2
HUM	19ENG232	Insights into Life through English Literature	2 0 0	2
HUM	19ENG233	Technical Communication	2 0 0	2
HUM	19ENG234	Indian Short Stories in English	2 0 0	2
HUM	19FRE230	Proficiency in French Language (Lower)	2 0 0	2
HUM	19FRE231	Proficiency in French Language (Higher)	2 0 0	2
HUM	19GER230	German for Beginners I	2 0 0	2
HUM	19GER231	German for Beginners II	2 0 0	2
HUM	19GER232	Proficiency in German Language (Lower)	2 0 0	2
HUM	19GER233	Proficiency in German Language (Higher)	2 0 0	2

HUM	19HIN101	Hindi I	2 0 0	2
HUM	19HIN111	Hindi II	2 0 0	2
HUM	19HUM230	Emotional Intelligence	2 0 0	2
HUM	19HUM231	Glimpses into the Indian Mind - the Growth of Modern India	2 0 0	2
HUM	19HUM232	Glimpses of Eternal India	2 0 0	2
HUM	19HUM233	Glimpses of Indian Economy and Polity	2 0 0	2
HUM	19HUM234	Health and Lifestyle	2 0 0	2
HUM	19HUM235	Indian Classics for the Twenty-first Century	2 0 0	2
HUM	19HUM236	Introduction to India Studies	2 0 0	2
HUM	19HUM237	Introduction to Sanskrit Language and Literature	2 0 0	2
HUM	19HUM238	National Service Scheme	2 0 0	2
HUM	19HUM239	Psychology for Effective Living	2 0 0	2
HUM	19HUM240	Psychology for Engineers	2 0 0	2
HUM	19HUM241	Science and Society - An Indian Perspective	2 0 0	2
HUM	19HUM242	The Message of Bhagwad Gita	2 0 0	2
HUM	19HUM243	The Message of the Upanishads	2 0 0	2
HUM	19HUM244	Understanding Science of Food and Nutrition	2 0 0	2
HUM	19JAP230	Proficiency in Japanese Language (Lower)	2 0 0	2
HUM	19JAP2313	Proficiency in Japanese Language (Higher)	2 0 0	2
HUM	19KAN101	Kannada I	2 0 0	2
HUM	19KAN111	Kannada II	2 0 0	2
HUM	19MAL101	Malayalam I	2 0 0	2
HUM	19MAL111	Malayalam II	2 0 0	2
HUM	19SAN101	Sanskrit I	2 0 0	2
HUM	19SAN111	Sanskrit II	2 0 0	2
HUM	19SWK230	Corporate Social Responsibility	2 0 0	2
HUM	19SWK231	Workplace Mental Health	2 0 0	2
HUM	19TAM101	Tamil I	2 0 0	2
HUM	19TAM111	Tamil II	2 0 0	2